



MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Master of Analytics (Business)

Paper Code: 115.801

Hydro-Risk Price Forecasting Model
Regime-Sensitive Assessment of Wholesale
Electricity Price Risk in New Zealand

Student Name: Siddharthan Muthukumaran

Student ID: 25007430

Supervisor: Dr. Shafiq Alam

Organisation: Wellington Electricity

Submission date: 21-Feb-2026

Table of Contents

MANAGERIAL ABSTRACT	1
INTRODUCTION	2
1.1 Research Topic	2
1.2 Decision-Making Problem	2
1.3 Importance of the Research	3
1.4 Contribution to the Company	3
1.5 Scope and Limitations	3
1.6 Structure of the Report	3
DESK RESEARCH / LITERATURE REVIEW	4
2.1 Introduction and Context	4
2.3 Academic Insights on Hydro Scarcity and Wholesale Price Risk	6
2.3.1 Electricity Price Formation in Hydro-Dominated Markets	6
2.3.2 Non-Linear and Regime-Based Price Behaviour	6
2.3.3 Volatility, Tail Risk, and Extreme Events in Electricity Markets	7
2.3.4 Limitations of Existing Forecasting and Risk Approaches	8
2.4 Industry and Practitioner Evidence	9
2.5 Synthesis and Research Gap	10
RESEARCH METHODOLOGY	12
3.1 Alternative Methodologies Considered	12
3.2 Selected Methodology and Justification	13
3.3 Analytical Workflow and Research Design	14
3.3.1 Data Sources	14
3.3.2 Data Preparation and Feature Engineering	14
3.3.3 Hydro-Risk and Price Analysis	15
3.3.4 Risk Interpretation	15
3.3.5 Decision Support Outputs	15
3.4 Ethical Considerations	16
3.5 Methodological Limitations and Critical Review	16
DATA ANALYSIS	18
4.1 Overview of Data and Sample Coverage	18
4.2 Visual Summary of Empirical Data	19

4.3 Analysis of the Data	21
4.3.1 Hydro Risk Regimes	21
4.3.2 Price Response under Dry-Year Conditions	23
4.3.3 Tail Behaviour and Extreme Events.....	24
4.3.4 Statistical Modelling and Quantitative Validation	26
4.4 Explanation of Major Findings / Insights	29
4.4.1 Hydro Scarcity as a Regime-Switching Mechanism	29
4.4.2 Tail Risk Amplification Under Scarcity.....	29
4.4.3 Dynamic and Intertemporal Market Adjustment	29
4.4.4 Structural Role of Hydro Storage in a Multi-Driver System	30
Concluding Synthesis.....	30
DISCUSSION AND DECISION IMPLICATIONS	31
5.1 Re-examining the Decision Problem, considering Empirical Findings.....	31
5.2 Correlation of Findings with Previous Research	32
5.3 Managerial and Policy Implications	32
5.3.1 Storage as a Risk Escalation Trigger	32
5.3.2 CVaR-Based Exposure Management.....	33
5.3.3 Dynamic Hedging and Persistence Awareness	33
5.3.4 Climate Adaptation and Long-Horizon Planning (2030–2050).....	33
5.3.5 Competitive and Strategic Positioning.....	34
Strategic Implication Summary	34
5.4 Critique of the Research.....	34
5.4.1 Model Scope and Structural Simplification.....	34
5.4.2 Regime Classification and Threshold Sensitivity	35
5.4.3 Temporal Aggregation Effects	35
5.4.4 Historical Dependence and Structural Change Risk.....	35
5.4.5 Omitted Climate Feedback Complexity.....	35
Critical Evaluation Summary.....	36
5.5 Further Research	36
5.5.1 Endogenous Regime-Switching Models.....	36
5.5.2 High-Frequency and Intraday Risk Modelling	36
5.5.3 Climate-Adjusted Hydrological Scenarios (2030–2050).....	37
5.5.4 Structural Market Evolution and Renewable Integration	37

5.5.5 Firm-Level Exposure and Hedging Effectiveness	37
5.5.6 Comparative Hydro-Dominated Markets	38
Strategic Research Outlook.....	38
CONCLUSION.....	39
6.1 Summary of Key Findings	39
6.2 Contribution to Knowledge.....	39
6.3 Contribution to Practice and Industry	40
6.4 Limitations of the Study.....	40
6.5 Directions for Future Research	41
6.6 Final Reflection.....	41
REFERENCES	42
APPENDIX.....	i
APPENDIX A — Data Inventory and Source Register	i
APPENDIX B — Dataset Validation and Lock Procedures	ii
APPENDIX C — Variable Dictionary	v
APPENDIX D — Low-Storage Stress Episodes	v
APPENDIX E — Robustness and Supplementary Modelling	vi
APPENDIX F — Supplementary Exploratory Visualisations	viii
APPENDIX G — Diagnostics	xx
APPENDIX H — Reproducibility and Code Extracts	xxiv

MANAGERIAL ABSTRACT

This study develops a regime-sensitive analytical framework to assess how hydro scarcity influences wholesale electricity price risk in New Zealand's hydro-dominated electricity market. Given the structural reliance on hydro generation, declining storage levels during dry-year conditions expose electricity retailers to significant volatility in procurement costs. The central objective is to determine whether hydro scarcity alters price behaviour structurally and to translate those effects into actionable risk-management signals.

Empirical results demonstrate that hydro storage operates as a regime-switching state variable. As storage approaches scarcity thresholds, price sensitivity intensifies non-linearly, upper-tail risk escalates materially, and elevated prices exhibit strong intertemporal persistence. Conditional Value-at-Risk (CVaR) metrics indicate that extreme price exposure increases substantially under low-storage conditions relative to standard regimes, confirming that hydro risk is asymmetric, tail-driven, and multi-period rather than gradual and mean-reverting.

The analysis is based on historical monthly data and does not explicitly model generator bidding behaviour or real-time market microstructure. Nevertheless, the results provide robust evidence of regime-dependent risk amplification under hydro stress.

For Wellington Energy, the findings imply that hydro storage thresholds should function as formal hedge escalation triggers. Risk monitoring should prioritise regime detection and tail-sensitive metrics rather than average price forecasts alone. The framework supports proactive procurement timing, persistence-aware hedging strategies, and climate-adjusted stress testing. By linking hydrological indicators directly to probabilistic risk measures, the study bridges hydro-economic theory and operational risk management within a hydro-dominated electricity market.

INTRODUCTION

1.1 Research Topic

Despite its heavy reliance on hydropower, New Zealand's wholesale electricity prices shift noticeably when water levels drop. When reservoirs run low, or rainfall falls short - so-called dry years - prices often swing sharply upward. Such conditions tend to trigger steep cost surges across the grid. Market actors like retailers face heightened exposure during these stretches, especially if buying power at spot rates, unpredictable swings in supply feed directly into fiscal uncertainty for those holding open positions.

This work develops a practical method to analyse how a limited water supply affects wholesale power prices. Using combined data on river flow, grid operations, and trading activity, it seeks to measure shifts in pricing patterns under different conditions. Early warnings tied to low-water periods are explored alongside their impact on financial exposure. Emphasis lies in turning physical water risks into clear indicators that inform economic decisions - specifically inside New Zealand's energy marketplace.

Hydro conditions gain greater significance because of how New Zealand's power market is structured. Given that a large share of electricity comes from hydropower, changes in water stored behind dams affect which plants set prices across the grid. When rain remains below average for extended periods, gas stations must run more often - these cost more to operate, squeezing available supply and pushing prices upward unpredictably. In such times, limited hydro availability stops being just one factor among many - it shapes the core pattern of how wholesale prices emerge

1.2 Decision-Making Problem

Electricity market participants in New Zealand face the ongoing challenge of managing exposure to volatile wholesale electricity prices, particularly during periods of hydro scarcity when price spikes can materially impact procurement costs and risk profiles. Existing decision-making approaches often rely on historical averages, short-term forecasts, or qualitative assessments of hydro conditions, which may fail to capture non-linear price responses and tail risk during extreme events. The central decision-making problem addressed in this research is how to identify emerging hydro-risk conditions early, quantify the associated wholesale price risk, and translate these insights into actionable operational and risk management decisions.

In particular, the challenge lies not only in recognising hydro scarcity once prices have already escalated, but in detecting early signals of regime transition before extreme price outcomes materialise. Without systematic identification of such transitions, risk management responses are likely to be reactive rather than proactive.

1.3 Importance of the Research

Understanding how water levels affect power prices is crucial for participants in New Zealand's electricity market. When dry years are not correctly forecast, buyers may face higher costs and weaker hedging plans because supply tightens under pressure. Better predictions of water-related risks help keep markets steady, as actors adjust early rather than scrambling later when shortages appear. Thus, this study advances practical tools for measuring risk in power networks reliant on hydropower, offering sharper insights for managing uncertainty effectively.

1.4 Contribution to the Company

This project contributes to practice within the New Zealand electricity market by delivering a structured, data-driven framework for monitoring hydro-related wholesale price risk and identifying early warning signals of adverse market conditions. By integrating hydro storage levels, inflows, spill volumes, HVDC flows, and climate indicators into a unified analytical pipeline, the research moves beyond descriptive monitoring to deliver quantifiable, decision-relevant risk intelligence. The project's outputs are designed to support practical decisions, such as procurement timing, risk escalation thresholds, and scenario-based planning during dry-year conditions.

1.5 Scope and Limitations

Hydro conditions shape how power prices move across New Zealand's electricity marketplace. When dry years are underestimated, buyers often face higher supply expenses. Instead of preparing early, some rely on delayed responses, which worsens hedge performance under pressure. Unexpected shortages become more complicated to manage without forward-looking tools. Better forecasts allow traders and planners to adjust before strain hits. Stability grows when decisions stem from insight rather than urgency. Analytical improvements help clarify the decoding of water-related risks. These methods support sharper judgment in complex energy environments. The work adds practical value to risk assessment when hydropower is centrally operated.

1.6 Structure of the Report

This report is structured as follows. Chapter 2 reviews relevant academic and practitioner literature to establish the conceptual foundations of hydro-related price risk and identify key research gaps. Chapter 3 outlines the methodological approach adopted to address the decision problem, including the analytical workflow and the justification for the selected techniques. Chapter 4 presents the empirical analysis, examining regime-dependent behaviour, dry-year persistence, and tail risk amplification in wholesale electricity prices. Chapter 5 discusses the implications of these findings for the decision problem and existing research, while critically reflecting on methodological limitations. Chapter 6 concludes by summarising the key insights and outlining the study's original contribution.

DESK RESEARCH / LITERATURE REVIEW

2.1 Introduction and Context

This chapter establishes the conceptual and empirical foundations underpinning the relationship between hydroelectric conditions and wholesale electricity price risk in hydro-dominated markets. The purpose of the desk research is not merely to summarise prior studies, but to synthesise key academic and practitioner insights relevant to understanding how hydro scarcity influences price dynamics and how such insights can inform decision-making within the New Zealand electricity market.

Hydro-dominated systems present a distinctive pricing environment in which water availability functions as both a physical constraint and an economic signal. Unlike fully thermal systems, where fuel inputs largely determine marginal costs, hydro systems introduce intertemporal decision-making through reservoir management and water opportunity costs. As a result, wholesale electricity prices in such markets are shaped not only by immediate supply-demand balance, but also by expectations of future inflows, storage trajectories, and scarcity risk. This creates conditions under which price behaviour may exhibit threshold effects, persistence, and asymmetry, particularly during sustained low-storage periods.

The literature relevant to this research can be broadly grouped into three interrelated streams. First, economic and market-based studies examining electricity price formation in hydro-dominated systems. Second, risk-focused research analysing volatility, tail behaviour, and extreme price events in electricity markets. Third, practitioner and industry-oriented work addressing operational risk monitoring and early warning indicators during dry-year events. By synthesising these streams, this chapter identifies key theoretical insights and practical limitations, ultimately motivating the need for an integrated, regime-sensitive analytical framework.

The chapter proceeds by outlining the context of the New Zealand electricity market, reviewing key academic contributions, examining practitioner evidence, and concluding with a synthesis that articulates the research gap this study addresses.

2.2 Industry and Market Context: The New Zealand Electricity System

The New Zealand electricity system is characterised by a high level of renewable generation, with hydroelectric power accounting for a substantial share of total electricity supply. This structural dependence on hydro generation distinguishes New Zealand from predominantly thermal systems and introduces unique dynamics into wholesale price formation. Because hydroelectric generation relies on reservoir storage and inflow conditions, water availability effectively functions as a strategic resource that shapes both short-term operational decisions and medium-term pricing expectations.

The wholesale electricity market in New Zealand operates under a marginal pricing framework, in which the market-clearing price is set by the most expensive generation unit required to meet demand in each trading period (Electricity Authority, 2023). Under normal hydrological conditions, hydro generation often supplies a substantial share of demand at relatively low marginal cost. However, as storage levels decline and thermal generation becomes increasingly necessary, the marginal unit shifts up the cost stack, resulting in higher wholesale prices. This pricing mechanism amplifies the economic consequences of hydro scarcity, as reductions in low-cost hydro supply directly translate into higher marginal prices.

Wellington Energy, as a market participant exposed to wholesale electricity prices, operates within this hydro-dominated pricing environment. Although retail pricing structures and contractual arrangements may buffer short-term fluctuations, sustained wholesale price increases during dry-year periods can materially affect procurement costs, hedging performance, and budget forecasting accuracy. Consequently, hydro-related price risk is not merely a theoretical concern but a direct financial exposure for the company.

Hydroelectric stations operate within physical storage constraints, meaning that generation decisions are inherently intertemporal. Water released for generation today cannot be recovered tomorrow and therefore carries an opportunity cost linked to expected future inflows and demand conditions during periods of sustained low inflows, reservoir levels decline, reducing available generation flexibility and increasing reliance on higher-cost thermal generation sources. As the marginal cost of supply rises, wholesale electricity prices may increase sharply, particularly if scarcity conditions persist.

Inter-island transmission dynamics further shape wholesale price formation. The High Voltage Direct Current (HVDC) link enables energy transfer between the hydro-dominant South Island and demand centres in the North Island. During periods of low hydro availability or transmission constraints, regional imbalances may intensify price volatility (Transpower, 2023). These structural transmission characteristics amplify the market-wide effects of hydro scarcity.

For Wellington Energy, such price escalations translate into elevated wholesale procurement costs, increased volatility in hedging positions, and heightened uncertainty in contract negotiations. Therefore, understanding how hydro scarcity affects both average prices and extreme outcomes is central to effective risk management within the company.

2.3 Academic Insights on Hydro Scarcity and Wholesale Price Risk

2.3.1 Electricity Price Formation in Hydro-Dominated Markets

Wholesale electricity price formation in hydro-dominated markets differs fundamentally from thermal systems due to the intertemporal nature of water as a generation resource. Stored water represents a strategic asset with an opportunity cost reflecting expected future scarcity. Unlike thermal systems, where marginal costs are directly linked to fuel inputs, hydro-based pricing incorporates expectations of inflows, reservoir trajectories, and forward-looking scarcity risk.

Economic theory conceptualises hydro storage as carrying an opportunity cost that rises as storage declines and uncertainty increases (Borenstein et al., 2002; Amundsen & Bergman, 2006). Empirical studies of hydro-dominated systems demonstrate that storage levels exert a non-linear influence on spot prices, with price sensitivity intensifying at lower reservoir levels (Bunn & Gianfreda, 2010; Lucia & Schwartz, 2002).

These findings imply that price–storage relationships are regime-dependent rather than stable across all hydrological states. In the New Zealand context, where hydro generation constitutes a substantial share of supply, storage levels function as a foundational structural determinant of wholesale price behaviour.

2.3.2 Non-Linear and Regime-Based Price Behaviour

Wholesale electricity price formation in hydro-dominated markets differs fundamentally from thermal systems due to the intertemporal nature of water as a generation resource. Stored water represents a strategic asset with an opportunity cost reflecting expected future scarcity. Unlike thermal systems, where marginal costs are directly linked to fuel inputs, hydro-based pricing incorporates expectations of inflows, reservoir trajectories, and forward-looking scarcity risk.

Economic theory conceptualises hydro storage as carrying an opportunity cost that rises as storage declines and uncertainty increases (Borenstein et al., 2002; Amundsen & Bergman, 2006). Empirical studies of hydro-dominated systems demonstrate that storage levels exert a non-linear influence on spot prices, with price sensitivity intensifying at lower reservoir levels (Bunn & Gianfreda, 2010; Lucia & Schwartz, 2002).

These findings imply that price–storage relationships are regime-dependent rather than stable across all hydrological states. In the New Zealand context, where hydro generation constitutes a substantial share of supply, storage levels function as a foundational structural determinant of wholesale price behaviour.

2.3.3 Volatility, Tail Risk, and Extreme Events in Electricity Markets

Electricity markets are widely recognised as exhibiting heavy-tailed distributions, price spikes, and volatility clustering (Weron, 2014; Knittel & Roberts, 2005). Unlike many financial assets, electricity prices are characterised by asymmetric distributions and extreme upper-tail behaviour driven by supply constraints and inelastic demand. Consequently, traditional variance-based risk measures may understate exposure, particularly during scarcity events. The application of tail-sensitive measures such as Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) has therefore become increasingly prominent in energy risk management research (Jorion, 2007; Rockafellar & Uryasev, 2000).

Traditional forecasting approaches often focus on expected or mean prices; however, the literature increasingly emphasises that mean-based measures are insufficient for capturing risk in electricity markets. Because price distributions are frequently skewed and characterised by extreme upper-tail outcomes, risk exposure is often driven by rare but severe price spikes rather than by gradual shifts in average levels. This has motivated the application of tail-sensitive risk measures such as Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR), which explicitly quantify the magnitude of losses associated with extreme percentiles of the distribution.

Studies examining electricity price risk highlight fat tails and asymmetric distributions, especially during periods of supply stress. Volatility clustering—where large price movements tend to be followed by further large movements—has also been observed in many electricity markets. In hydro-based systems, such clustering may be linked to sustained scarcity conditions, where declining storage levels create persistent upward pressure on prices. As scarcity intensifies, the probability of extreme price events increases, thickening the distribution's upper tail.

Notably, several researchers argue that extreme price behaviour in electricity markets is not merely random noise but reflects the market's structural characteristics. Limited storage capability for electricity, inelastic short-term demand, and capacity constraints combine to produce occasional price spikes when supply margins tighten. In hydro-dominated systems, reservoir depletion amplifies these mechanisms by reducing low-cost supply and increasing dependence on more expensive generation sources. Consequently, extreme price events may be interpreted as manifestations of system-level stress rather than isolated anomalies.

The literature, therefore, advocates risk frameworks that explicitly incorporate tail behaviour rather than relying solely on average volatility measures. While standard deviation captures overall dispersion, it does not adequately represent the severity of outcomes once extreme thresholds are breached. Conditional tail measures, such as CVaR, provide a more informative assessment of downside exposure by estimating the expected loss magnitude at the worst-case percentiles.

In the context of hydro scarcity, these insights imply that risk assessment must extend beyond central price forecasts to include evaluation of distributional shifts and upper-tail amplification. The empirical findings presented in Chapter 4, which demonstrate substantial divergence in high-percentile prices and tail metrics between standard and low-storage regimes, align with this body of literature. By explicitly quantifying tail risk, the analytical framework adopted in

this study addresses a core limitation identified in prior research and enhances the decision relevance of wholesale price risk assessment.

2.3.4 Limitations of Existing Forecasting and Risk Approaches

Despite the substantial body of research examining hydro-driven price dynamics and extreme outcomes in electricity markets, several methodological limitations persist within the existing literature. A central concern is the continued reliance on models that assume stable relationships across time and hydrological conditions. Linear regression frameworks calibrated on long historical samples may perform adequately under average conditions but often underestimate risk during low-storage regimes. By averaging across structurally different system states, such a model obscures threshold effects and regime transitions that are critical for accurate risk assessment.

Similarly, many forecasting approaches prioritise point predictions of mean prices rather than distributional characteristics. While mean forecasts may provide practical baseline expectations, they offer limited insight into the probability and magnitude of extreme price outcomes. In markets characterised by heavy-tailed distributions and scarcity-driven shocks, risk exposure is disproportionately influenced by upper-tail events rather than by gradual changes in central tendencies. Consequently, frameworks that do not explicitly incorporate tail-sensitive measures risk underrepresenting the accurate scale of exposure during stress periods.

Another limitation concerns the gap between theoretical modelling and operational application. A considerable portion of the academic literature focuses on structural market models or advanced econometric specifications that, while analytically rigorous, may lack transparency or practical interpretability. In applied settings, decision-makers require indicators that are not only statistically robust but also easily monitored, understood, and acted upon. Complex models that obscure causal mechanisms or produce opaque outputs may therefore face challenges in real-world implementation.

From a practitioner perspective, risk monitoring often relies on isolated indicators such as storage levels or inflow projections assessed independently. While such metrics provide valuable signals, they are rarely integrated into unified frameworks that quantify how multiple hydro-related factors jointly influence price risk. This fragmentation limits the ability to systematically translate hydro conditions into probabilistic assessments of price escalation and extreme events.

These limitations highlight the need for analytical approaches that combine regime identification, distributional risk measurement, and interpretability within a coherent framework. By integrating hydrological indicators with tail-focused risk metrics and explicitly distinguishing between normal and scarcity regimes, the present research seeks to address these methodological gaps. In doing so, it moves beyond average-based forecasting toward a structured assessment of hydro-related price risk that is both empirically grounded and decision-relevant.

2.4 Industry and Practitioner Evidence

Practitioner-oriented literature and industry reporting provide complementary insight into the real-world implications of hydro scarcity within the New Zealand electricity market. System operators, regulators, and market analysts consistently recognise dry-year conditions as periods of heightened risk, characterised by declining reservoirs, increased reliance on thermal generation, and elevated wholesale price volatility.

Dry-year risk has also been recognised at the policy level in New Zealand's electricity market design. Government reviews and energy strategy documents highlight hydro inflow variability as a structural source of security-of-supply risk, particularly during prolonged low-inflow events (Ministry of Business, Innovation and Employment [MBIE], 2022). These policy discussions reinforce the systemic importance of early identification of hydro stress conditions.

Transpower's Security of Supply assessments emphasise the importance of hydro storage trajectories and inflow expectations in managing dry-year risk, noting that sustained low storage increases the probability of supply stress and elevated market prices (Transpower, 2022). These assessments reinforce the structural linkage between hydrological conditions and wholesale price volatility.

For Wellington Energy, such periods represent not only market-wide stress but also measurable financial exposure. Wholesale procurement strategies typically rely on a mix of spot purchases and hedging contracts. During dry-year conditions, elevated and persistent spot prices can push procurement costs above hedged levels, while rising forward prices may raise the cost of new hedge contracts. Consequently, hydro scarcity introduces both realised and anticipatory cost risk.

Industry practice often relies on monitoring storage percentages relative to historical averages, inflow projections, and seasonal outlooks. However, these indicators are frequently assessed in isolation rather than integrated into probabilistic risk frameworks. For a company seeking to optimise hedge timing and reduce volatility exposure, qualitative monitoring may be insufficient. Instead, there is a need for quantitative systems that translate hydro indicators into measurable price-risk thresholds and automated escalation triggers.

This gap between qualitative monitoring and structured risk quantification is particularly relevant to the objectives of the Hydro-Risk Price Forecasting Model. While hydro scarcity is widely acknowledged as a driver of price volatility, fewer practitioner frameworks explicitly quantify tail risk or identify regime transitions in a statistically defensible manner.

Market statistics and historical pricing data published by the Electricity Authority through the Electricity Market Information (EMI) platform further document the relationship between hydro conditions and wholesale price variability (Electricity Authority, 2023). However, these datasets are typically used descriptively rather than within integrated probabilistic risk frameworks.

2.5 Synthesis and Research Gap

What we find across studies points clearly to water shortage shaping how power prices move in places like New Zealand, where hydropower runs the grid. Instead of treating water as just another input, economics sees it as a resource with future value - using it now means giving up later gains, which nudges prices upward. Data-backed work shows prices do not react smoothly; they shift sharply under stress, especially when reservoirs run low. When drought hits, swings grow wilder, patterns break, and rare spikes dominate total risk more than expected. Market reports back this up - dry seasons strain both operations and balance sheets, proving real-world weight behind theoretical ideas.

Even though theory and practice have moved closer together, key shortcomings still exist. One issue stands out: scholarly work tends to study water-related price shifts separately from how decisions are made in real operations. Although complex patterns and rare events are widely recognised, only a small number of analyses turn such insights into organised tracking tools or timely alerts suited to traders and operators. Another limitation appears in predictions, which typically prioritise average prices instead of mapping full risk distributions - an approach that falls short when severe high-end scenarios define vulnerability. A third gap emerges in field practices, where professionals often watch single metrics like reservoir levels or rainfall estimates without combining various hydrologic and operational factors within one coherent system for evaluating danger.

Although the core drivers of hydropower-linked price risks are generally known, turning those insights into practical tools has yet to be fully achieved. Climate fluctuations and gradual changes in water availability could heighten market volatility, as newer studies point out (IEA, 2022). Because of this, models used to assess pricing risks ought to include updated climate-informed water forecasts. Where hydropower shapes most of the supply, managing risk well goes beyond noticing lower reservoir volumes. Instead, it involves detecting shifts in system behavior early, measuring how extreme outcomes become more likely, then directly tying water data to statistical estimates of price surges.

This study tackles existing shortcomings through creation of a connected analysis system, responsive to shifting conditions, merging water flow data, grid operations, and pricing trends to measure power cost volatility during periods of limited hydropower supply. Rather than relying on general patterns, the method separates typical storage phases from critically low ones while emphasizing downside risks, focusing attention where impacts are most severe. Through such differentiation, insights drawn from theory align more closely with real-world planning needs, especially within markets shaped heavily by renewable generation fluctuations. Its design responds directly to how New Zealand's energy network behaves when water reserves tighten.

What studies show points toward drought-driven electricity prices acting unpredictably - shifting sharply under certain thresholds. Yet current methods tend to stay trapped in abstract theory or fall back on informal observation lacking structured risk analysis. In practice, Wellington Energy faces uncertainty: even when water levels are known, their influence on severe market pricing stays unclear across different states of system behaviour. Bridging this

demands an approach attentive to changing regimes, one centred on full probability distributions rather than isolated forecasts. Building such a tool forms the core aim here.

RESEARCH METHODOLOGY

3.1 Alternative Methodologies Considered

A range of methodological approaches have been employed in the academic literature to model electricity price dynamics and risk. These approaches can broadly be categorised into time-series models, structural market models, and machine learning frameworks. Each offers distinct advantages, yet each presents limitations when applied to the analysis of hydro-driven price risk in a hydro-dominated electricity system such as New Zealand.

Traditional time-series models, including autoregressive integrated moving average (ARIMA) specifications and volatility models such as GARCH, are widely used to capture short-term price dynamics and volatility clustering. These models are effective in representing temporal dependencies and conditional variance behaviour. However, they typically assume stable relationships over time and may struggle to capture regime shifts associated with structural changes in hydro storage conditions. In hydro-dominated markets where price behaviour differs substantially between regular and low-storage periods, purely time-series approaches may understate risk during scarcity regimes by averaging across heterogeneous system states.

Structural market models provide an alternative approach by explicitly modelling supply-demand fundamentals, generation costs, and bidding behaviour. These models aim to simulate marginal pricing mechanisms and capture the economic logic underpinning price formation. While theoretically appealing, structural models require detailed assumptions regarding generator cost structures, bidding strategies, and market microstructure. In practice, such detailed data are often unavailable or difficult to validate. Moreover, structural models may become highly complex, reducing transparency and interpretability for operational decision-making contexts.

More recently, machine learning methods—including tree-based models, neural networks, and ensemble techniques—have been applied to electricity price forecasting. These approaches offer flexibility in modelling non-linear relationships and variable interactions. However, they often operate as “black box” systems, limiting interpretability and making it challenging to attribute risk to specific hydro-related drivers. In applied business contexts where decision-makers require clear, explainable indicators, a lack of transparency may hinder adoption and trust in model outputs.

Given the objective of this research—to develop a regime-sensitive and decision-relevant framework for assessing hydro-related wholesale price risk—these alternative methodologies were considered but not adopted as the primary analytical approach. The limitations of purely linear time-series models, the complexity of structural simulations, and the opacity of advanced machine learning techniques motivate the selection of an applied, interpretable framework capable of capturing non-linear regime behaviour and tail risk while remaining operationally transparent.

3.2 Selected Methodology and Justification

This study adopts an applied, data-driven analytical framework that balances empirical rigour with interpretability and decision relevance. Rather than relying on a single forecasting technique, the methodology integrates hydrological, operational, and market variables within a regime-sensitive, tail-focused analytical framework. The objective is not merely to predict average wholesale electricity prices, but to identify hydro-related regime transitions and quantify associated extreme price risk.

The analytical approach is structured around three core components. First, hydro risk is conceptualised through observable hydrological indicators, including national hydro storage levels, tributary inflows, and spill volumes. These variables capture both the level and trajectory of water availability, which economic theory identifies as central to price formation in hydro-dominated systems. Operational variables, including HVDC inter-island flows, are incorporated to reflect transmission dynamics and spatial balancing within the electricity system. Climate indicators are also included to account for broader environmental conditions influencing inflows and system stress.

Second, the methodology explicitly incorporates regime identification. Drawing on the literature on non-linear, threshold-driven price behaviour, the analysis distinguishes between standard and low-storage regimes using storage-based classification rules. This regime-based structure allows empirical relationships and risk metrics to vary across system states rather than assuming stable dynamics across all hydrological conditions. By separating scarcity and non-scarcity environments, the framework captures structural differences in price behaviour that would otherwise be masked by aggregate modelling.

Third, the framework emphasises the measurement of distributional risk. Instead of focusing solely on mean price forecasts, the analysis incorporates tail-sensitive metrics, including high-percentile measures and conditional tail indicators. These metrics quantify the magnitude and severity of extreme price outcomes during dry years. The inclusion of such measures addresses a key limitation identified in the literature: the tendency of average-based models to underestimate exposure in markets with heavy tails.

Monthly aggregation is adopted as the primary temporal frequency to align hydrological indicators with observable price dynamics while smoothing short-term noise that may obscure structural relationships. Given the intertemporal nature of hydro storage and the cumulative development of dry-year conditions, monthly data provide an appropriate balance between granularity and structural clarity. This frequency supports the identification of regime transitions and persistence effects without overfitting short-term fluctuations.

The selected methodology prioritises transparency and interpretability. Each component—regime classification, distributional metrics, and variable integration—can be clearly communicated and monitored. This design ensures that outputs remain operationally meaningful while retaining analytical credibility. By aligning methodological choices with the research gap identified in Chapter 2, the framework moves beyond purely statistical modelling toward a structured assessment of hydro-related wholesale electricity price risk that is both empirically grounded and decision-relevant.

3.3 Analytical Workflow and Research Design

The analytical workflow adopted in this study is structured to ensure transparency, reproducibility, and logical progression from raw data to decision-relevant insights. Figure 3.1 summarises the sequence of stages through which multiple datasets are transformed into interpretable risk indicators. Each stage of the workflow serves a distinct purpose within the broader research design.

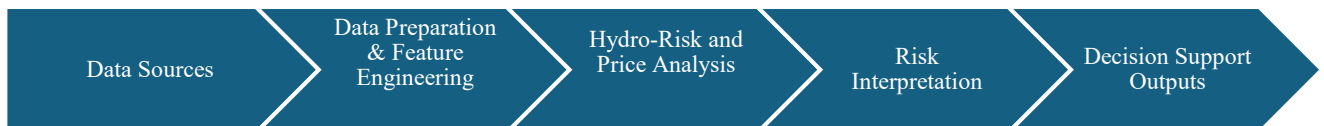


Figure 3.1: Analytical Workflow for Hydro-Risk Price Assessment

3.3.1 Data Sources

The analysis begins by integrating multiple datasets capturing the hydrological, operational, climatic, infrastructural, and market dimensions of the New Zealand electricity system. These include national hydro storage levels, tributary inflows, spill volumes, HVDC inter-island flows, climate indicators, and wholesale electricity prices. In addition, static infrastructure attributes are incorporated to contextualise system capacity and constraints.

The inclusion of diverse but complementary data sources ensures that hydro-related price risk is examined within the whole operational environment of the electricity system rather than in isolation.

3.3.2 Data Preparation and Feature Engineering

Raw datasets are cleaned, standardised, and harmonised to a consistent monthly frequency. A common Year–Month index is used to align variables across sources, ensuring temporal comparability. Data preparation includes handling missing values, validating units of measurement, and constructing derived indicators where required.

Feature engineering focuses on transforming raw hydrological and market variables into analytically meaningful indicators. This includes:

- Construction of storage-based regime classifications
- Calculation of rolling price metrics
- Derivation of volatility measures
- Creation of change indicators (e.g., storage differences)

These transformations enable the identification of regime transitions, persistence effects, and distributional risk characteristics.

3.3.3 Hydro-Risk and Price Analysis

The core empirical analysis examines relationships between hydro conditions and wholesale electricity price behaviour. This stage includes:

- Regime-based analysis comparing storage states
- Examination of non-linear relationships between storage and price
- Quantification of volatility and distributional shifts
- Tail-risk measurement using high-percentile and conditional metrics

Rather than relying on a single modelling technique, the research design combines descriptive, comparative, and distributional analysis to capture structural features of hydro-related price risk.

3.3.4 Risk Interpretation

Empirical results are interpreted through a risk-management lens. Particular emphasis is placed on identifying:

- Persistence of price elevation during dry-year periods
- Escalation of upper-tail outcomes under scarcity
- Differences between average effects and tail-driven exposure

This stage moves beyond statistical reporting to interpret the implications of the findings for risk monitoring and early warning design.

3.3.5 Decision Support Outputs

The final stage translates analytical findings into decision-relevant outputs. These include:

- Regime indicators
- Tail-risk metrics
- Threshold-based monitoring signals

The objective is to ensure that empirical insights are not confined to academic interpretation but can inform procurement timing, risk escalation thresholds, and stress-period planning within the New Zealand electricity market.

3.4 Ethical Considerations

This research relies exclusively on secondary data obtained from publicly available sources and institutionally provided datasets related to the New Zealand electricity market. No personal, confidential, or commercially sensitive individual-level data is used. All analysis is conducted at an aggregate market level, focusing on system-wide price behaviour and hydro conditions rather than firm-specific or customer-level information.

Data handling and analytical procedures were conducted in accordance with Massey University's ethical guidelines for research. Particular attention was given to transparency, reproducibility, and responsible data use. All data transformations, cleaning procedures, and analytical steps were implemented using documented code within a structured computational environment to ensure that results can be replicated and verified. This approach supports methodological integrity and reduces the risk of undocumented manipulation or selective reporting.

Although the study does not involve human participants and therefore does not raise issues related to informed consent or personal privacy, ethical responsibility remains relevant in the interpretation and communication of results. Electricity price risk analysis can influence perceptions of market stability and risk exposure. Accordingly, findings are presented with appropriate contextualisation, acknowledging uncertainty and methodological limitations rather than overstating predictive precision.

Finally, the research avoids speculative conclusions regarding individual market participants or strategic behaviour. The analysis is confined to observable system-level dynamics and empirical relationships, ensuring that conclusions remain evidence-based and analytically defensible.

3.5 Methodological Limitations and Critical Review

While the selected analytical framework is well aligned with the objectives of this research, several methodological limitations warrant consideration. First, the analysis focuses on system-level relationships between hydro conditions and wholesale electricity price behaviour, rather than explicitly modelling strategic generator bidding behaviour or detailed short-term market microstructure. As a result, the framework captures broad regime dynamics and tail-risk patterns but does not simulate the precise operational mechanisms by which individual market participants influence price formation.

Second, the identification of hydro risk regimes is based on historical storage thresholds and empirical distributional properties. Although this approach allows for transparent and interpretable regime classification, threshold selection may influence the precise magnitude of estimated risk measures. While sensitivity checks were undertaken to ensure robustness, regime boundaries remain an empirical simplification of a complex hydrological system.

Third, the analysis relies on historical data and assumes that past relationships between hydro conditions and wholesale prices provide meaningful guidance for future risk assessment. Structural changes in generation mix, regulatory settings, demand patterns, or transmission

constraints may alter these relationships over time. Consequently, results should be interpreted as conditional on historical market structure rather than as immutable behavioural laws.

Additionally, aggregating data to a monthly frequency enhances stability and interpretability but may smooth short-term volatility and intramonth dynamics. While appropriate for identifying persistent hydro scarcity regimes, this temporal aggregation may understate short-lived extreme price spikes that occur within monthly intervals.

Despite these limitations, the methodology offers several strengths. Its emphasis on regime identification, non-linear behaviour, and tail-sensitive metrics provides a transparent and operationally interpretable framework for assessing hydro-related price risk. By balancing empirical rigour with practical applicability, the approach generates insights that are both analytically defensible and decision-relevant. The limitations outlined above do not invalidate the findings but instead define the appropriate scope within which conclusions should be interpreted.

DATA ANALYSIS

4.1 Overview of Data and Sample Coverage

This chapter presents the empirical analysis of wholesale electricity price behaviour in relation to hydroelectric conditions in New Zealand. The analysis integrates multiple datasets capturing hydrological, operational, climatic, infrastructural, and market characteristics of the electricity system. All time-varying datasets are harmonised to a consistent monthly frequency and aligned using a common Year–Month index to ensure comparability across sources and analytical coherence.

The primary time-varying datasets include national hydro storage levels, tributary inflows, spill volumes, HVDC inter-island flows, climate indicators, and wholesale electricity prices. These datasets collectively capture both normal operating conditions and periods of hydro scarcity, enabling analysis of regime-dependent behaviour, non-linear price responses, and extreme outcomes associated with dry-year events.

In addition to dynamic variables, the analysis incorporates hydro infrastructure and constraint attributes to provide system-level context. These infrastructure variables describe the physical and operational characteristics of the hydro system and are treated as structural information rather than time-varying drivers in the empirical analysis. Their inclusion ensures that observed price dynamics are interpreted within the broader constraints of the New Zealand electricity system without introducing unnecessary complexity into the modelling framework.

The final analytical dataset spans multiple decades, covering a wide range of hydrological conditions and market environments. This extensive coverage supports robust inference across both average conditions and stress periods, forming a strong empirical foundation for the regime-based and risk-focused analysis that follows.

The full aligned dataset spans from 1996 to 2024, providing 279 monthly observations across the longest available series and capturing multiple hydrological cycles, including both standard and dry-year conditions. However, the econometric regression models reported in Section 4.3.4 are constrained by the availability of HVDC flow data and lag construction, resulting in a modelling sample spanning 1996–2019 (241 monthly observations). This distinction ensures that descriptive analysis reflects the most recent market behaviour, while regression results remain methodologically consistent across all included drivers.

Data Category	Key Variables	Frequency	Time Coverage	Source
Hydro infrastructure & constraints	Installed capacity, constraint attributes	Static	Not time-varying	Infrastructure dataset
Hydro storage	National storage (%)	Monthly	Full sample	Electricity Authority
Inflows	Tributary inflows	Monthly	Full sample	NIWA
Spill	Spill volumes	Monthly	Full sample	Generator reports
HVDC flows	Inter-island flows	Monthly	Full sample	Transpower
Climate	Rainfall/climate indices	Monthly	Full sample	NIWA
Prices	Wholesale electricity price (NZD/MWh)	Monthly	Full sample	Market data

Table 4.1 - Data Sources and Coverage

To ensure analytical comparability, all time-varying variables were standardised to a consistent monthly frequency and aligned using a common Year–Month index. Derived variables were constructed where appropriate, including rolling averages and volatility measures for wholesale prices, as well as first-difference transformations of storage to capture dynamic changes in hydro conditions. These engineered features allow the analysis to capture both level effects and momentum effects in hydro scarcity dynamics.

4.2 Visual Summary of Empirical Data

This section presents a concise visual overview of the core empirical variables underpinning the analysis to establish system-level intuition before undertaking a detailed analytical assessment. The figures are selected to illustrate the temporal behaviour of wholesale electricity prices and hydroelectric conditions, highlighting patterns of variability, persistence, and extremity that are central to hydro-related price risk. Rather than providing an exhaustive set of exploratory graphics, the visuals included in this section focus on the most informative features of the data that motivate a risk-focused, regime-based analytical approach in subsequent sections.

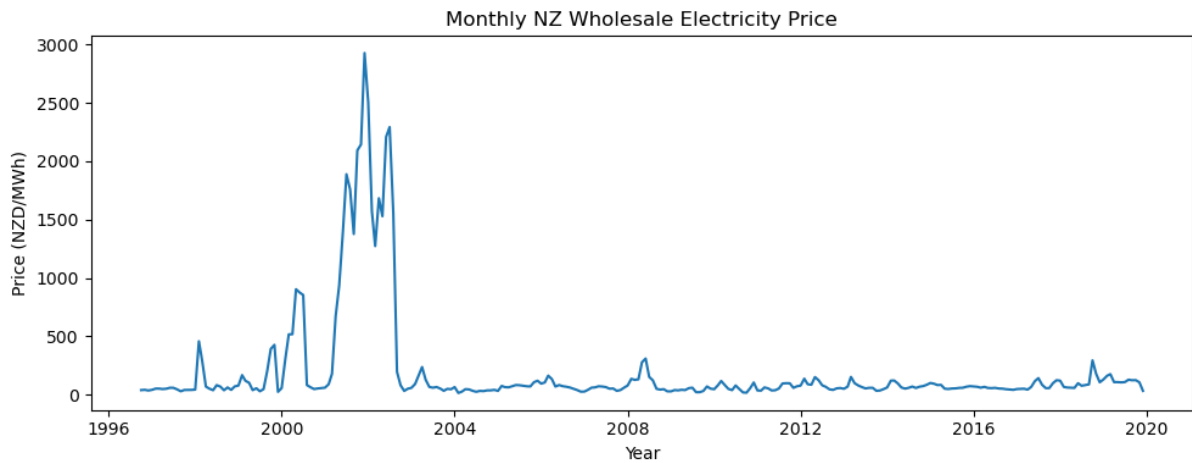


Figure 4.1a: Monthly Wholesale Electricity Prices in New Zealand

Figure 4.1a illustrates the evolution of monthly wholesale electricity prices over the whole sample period. Prices exhibit long stretches of relative stability punctuated by infrequent but extreme spikes. These spikes are large in magnitude compared to the central mass of observations and occur episodically rather than continuously. This behaviour suggests that price risk is highly concentrated in discrete stress periods rather than evenly distributed over time. The episodic nature of price spikes indicates that conventional models relying solely on central tendencies may understate exposure during extreme market conditions.

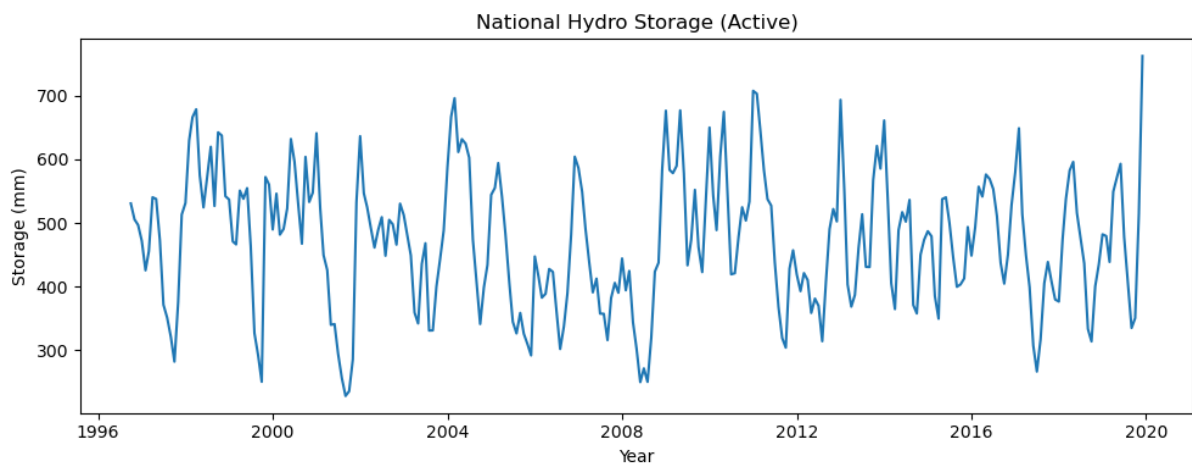


Figure 4.1b — National Hydro Storage (Active)

Figure 4.1b presents the trajectory of national hydro storage levels over the same period. In contrast to the sudden nature of price spikes, hydro storage evolves gradually and often declines cumulatively across multiple months. Periods of sustained inflow shortfall result in progressive depletion rather than abrupt shocks. This persistence implies that hydro scarcity develops over time and may provide warning signals of deteriorating system conditions before extreme price outcomes fully materialise.

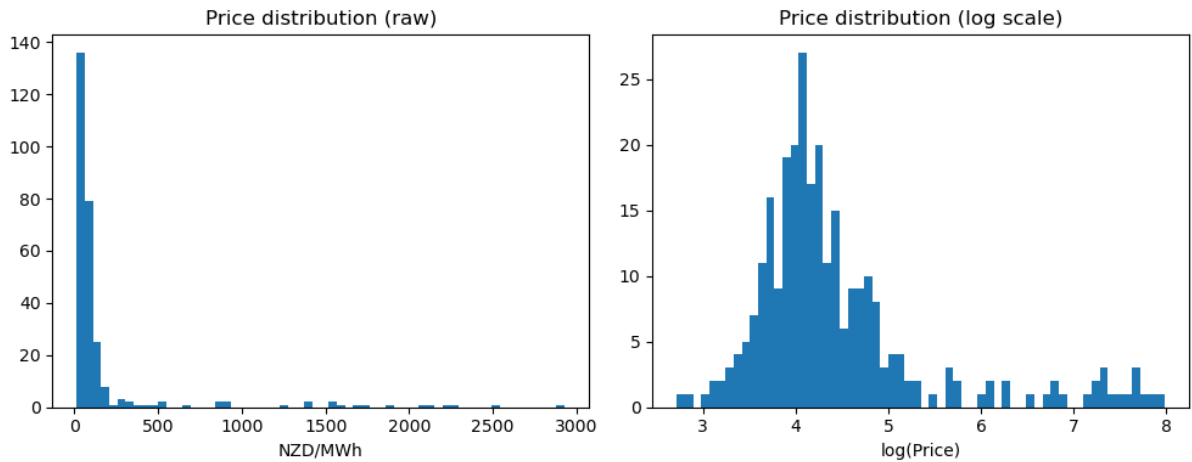


Figure 4.2 — Distribution of Wholesale Electricity Prices (Raw and Log Scale)

Figure 4.2 highlights the pronounced right-skewness of wholesale electricity prices. While most observations are concentrated at moderate levels, a small number of extreme values extend far into the upper tail. The log-transformed distribution appears substantially more symmetric, confirming that extreme outcomes dominate the level-based distribution. This asymmetry indicates that price risk is driven disproportionately by upper-tail events rather than uniform shifts in average prices. This motivates the use of regime-based and tail-sensitive metrics in Section 4.3.

Overall, the visual evidence presented in this section highlights three core empirical characteristics of the dataset: episodic price spikes, persistent hydro storage dynamics, and significant distributional asymmetry. Together, these features justify moving beyond linear, mean-focused modelling approaches and adopting a regime-based and tail-sensitive framework to assess hydro-related wholesale electricity price risk in the New Zealand market.

4.3 Analysis of the Data

This section presents the core empirical analysis examining how hydroelectric conditions influence wholesale electricity price behaviour in New Zealand. Building on the system-level patterns established in Section 4.2, the analysis moves beyond visual intuition to formally assess regime-dependent behaviour, non-linear dynamics, and risk amplification under hydro scarcity. The focus is not solely on average price effects, but also on how the entire price distribution shifts as storage conditions deteriorate. Particular attention is given to identifying threshold effects and structural changes in volatility and tail outcomes.

4.3.1 Hydro Risk Regimes

To evaluate whether wholesale electricity prices respond differently under varying hydro conditions, storage levels were categorised into distinct regimes based on empirical percentiles. This classification enables systematic comparison of price behaviour across hydrological states and facilitates formal assessment of non-linear dynamics. By segmenting the sample into high, normal, and low-storage regimes, the analysis moves beyond continuous linear modelling and allows for structural comparison of pricing behaviour under differing scarcity conditions.

Wholesale electricity prices exhibit clear regime-dependent characteristics. Under high-storage conditions, price levels remain comparatively contained, and volatility is moderate. In contrast, as storage declines into the low-storage regime, price variability increases substantially, and extreme outcomes become more frequent. This pattern suggests that hydro scarcity influences prices in a threshold-driven manner, where risk intensifies once storage falls below critical levels rather than increasing proportionally across the entire storage range.

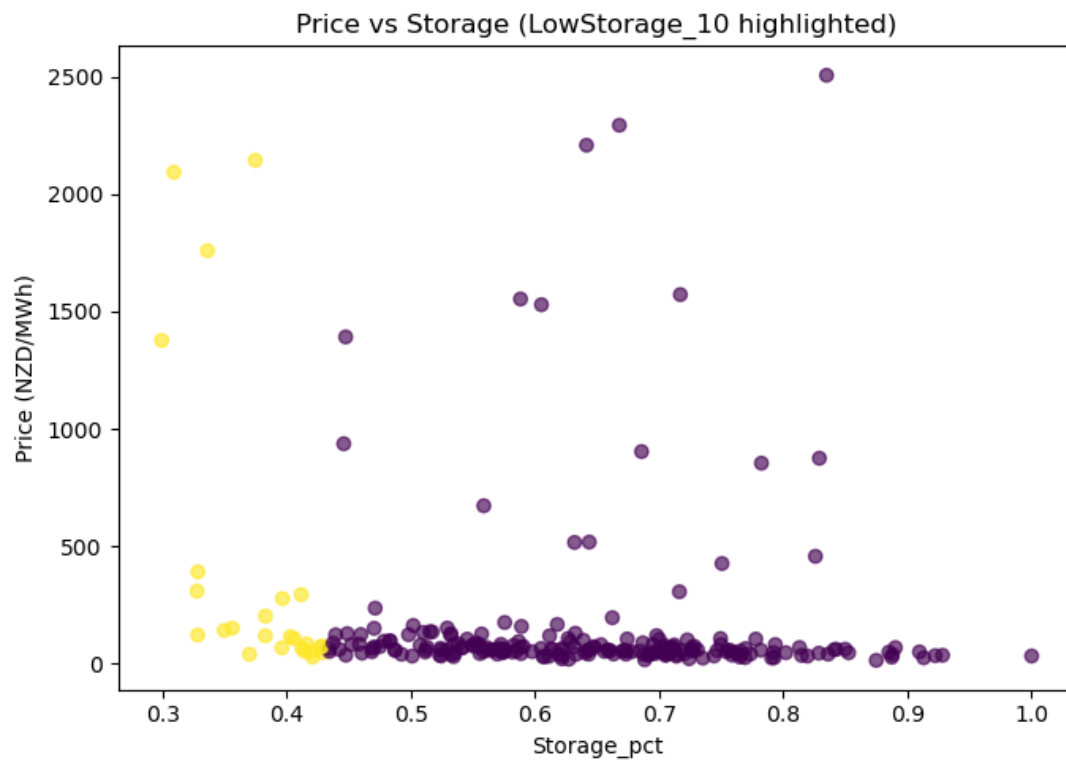


Figure 4.3: Relationship between hydro storage levels and wholesale electricity prices, highlighting the low-storage regime

Figure 4.3 visually illustrates this non-linear relationship. While prices remain relatively dispersed but bounded across moderate to high storage levels, extreme price observations are concentrated predominantly within the low-storage region. The clustering of high-price realisations at low storage levels provides clear evidence of regime-dependent behaviour. It supports the use of a threshold-based framework for assessing hydro-related price risk.

Storage_Regime	Mean_Price	Median_Price	Max_Price	Volatility	P90	P95	Months
High storage	140.97	50.46	2506.2	363.81	197.65	556.56	56
Low storage	300.96	95.73	2144.07	541.46	1156.8	1791.5	56
Normal storage	173.58	64.69	2927.9	412.48	170.51	834.14	167

Table 4.2: Summary statistics of wholesale electricity prices by hydro storage regime

The regime-based summary statistics quantify the magnitude of price behaviour divergence. Mean prices during low-storage periods (300.96 NZD/MWh) are materially higher than those observed under normal (173.58 NZD/MWh) and high-storage conditions (140.97 NZD/MWh). Volatility also increases significantly in the low-storage regime, reflecting greater dispersion and instability in pricing outcomes.

The divergence is particularly pronounced in upper-tail metrics. The 90th and 95th percentile prices in the low-storage regime exceed those in other regimes by a substantial margin, indicating that hydro scarcity disproportionately amplifies extreme outcomes rather than merely shifting average prices upward. This asymmetric escalation in upper-tail thresholds confirms that hydro scarcity alters the distributional structure of prices, reinforcing the limitations of linear, mean-based representations.

Overall, the structural separation between regimes provides strong empirical evidence that hydro conditions function as a systemic determinant of wholesale electricity price behaviour. These findings establish the foundation for the subsequent analysis of dry-year persistence and tail-risk amplification.

4.3.2 Price Response under Dry-Year Conditions

This subsection examines how wholesale electricity prices respond during sustained hydro scarcity, commonly referred to as dry-year conditions. Building on the regime-based evidence presented in Section 4.3.1, the focus here is on how price behaviour once storage enters the low-storage regime (defined as national storage below approximately **34% of active capacity**, corresponding to the lowest decile of historical observations). Attention is given to persistence, escalation, and distributional amplification relative to normal operating conditions. The objective is to distinguish between temporary price fluctuations and prolonged stress periods that materially affect market risk exposure.

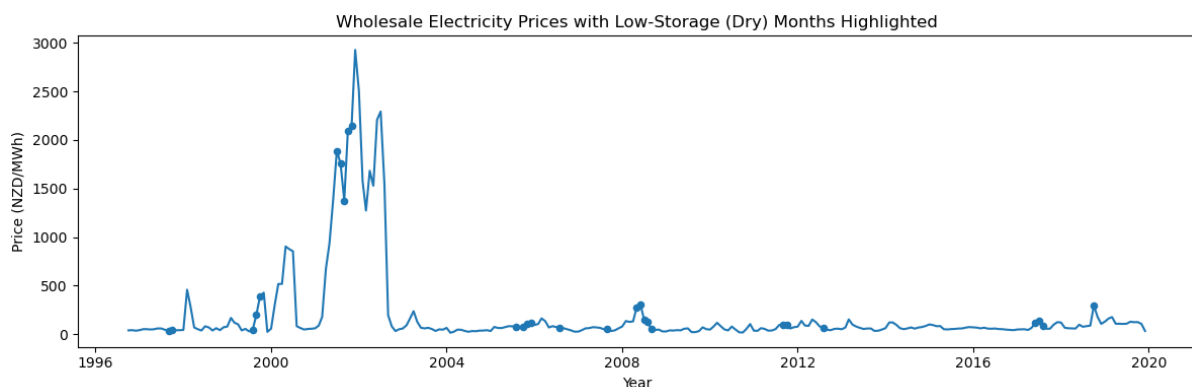


Figure 4.4: Wholesale electricity price behaviour during dry-year (low-storage) periods

Figure 4.4 shows that dry-year periods are characterised by sustained elevations in wholesale electricity prices rather than isolated, short-lived spikes. Once hydro storage falls into the low-storage regime, prices tend to remain elevated across multiple consecutive months. This persistence indicates cumulative system stress rather than transient volatility shocks. Unlike normal fluctuations, dry-year price behaviour reflects prolonged scarcity that limits the system's ability to revert quickly to baseline pricing dynamics. The temporal clustering of high-

price realisations suggests that hydro-related risk is multi-period in nature and therefore materially different from short-term market disturbances.

Period	Mean Price	Median Price	Max Price	Volatility	P90	P95	Months
Dry year (low storage)	438.16	117.04	2144.07	687.59	1798	2021.5	28
Normal	165.21	61.33	2927.9	389.22	175.51	864.53	251

Table 4.3: Comparison of wholesale electricity price behaviour during dry-year and regular periods

Dry-year periods are characterised by sustained price elevation rather than isolated spikes. Once hydro storage enters the low regime, elevated prices frequently persist across consecutive months, reflecting cumulative system stress rather than transient volatility.

Table 4.3 quantifies this divergence. Mean prices during dry-year months are more than double those observed under normal conditions, while volatility and upper-tail thresholds increase substantially. Although dry-year months account for only a small proportion of the sample, they contribute disproportionately to extreme price outcomes.

These findings confirm that hydro scarcity generates a structurally distinct and persistent risk regime, reinforcing the importance of early detection and proactive risk management.

4.3.3 Tail Behaviour and Extreme Events

This subsection examines wholesale electricity price behaviour in the extreme upper tail of the distribution, where financial exposure is most severe. While Sections 4.3.1 and 4.3.2 demonstrated that hydro scarcity generates regime-dependent and persistent price escalation, this section focuses on quantifying how dry-year conditions alter both the probability and severity of extreme price outcomes. The analysis, therefore, shifts from central tendency measures to tail-risk dynamics, which are critical for understanding exposure during sustained hydro stress.

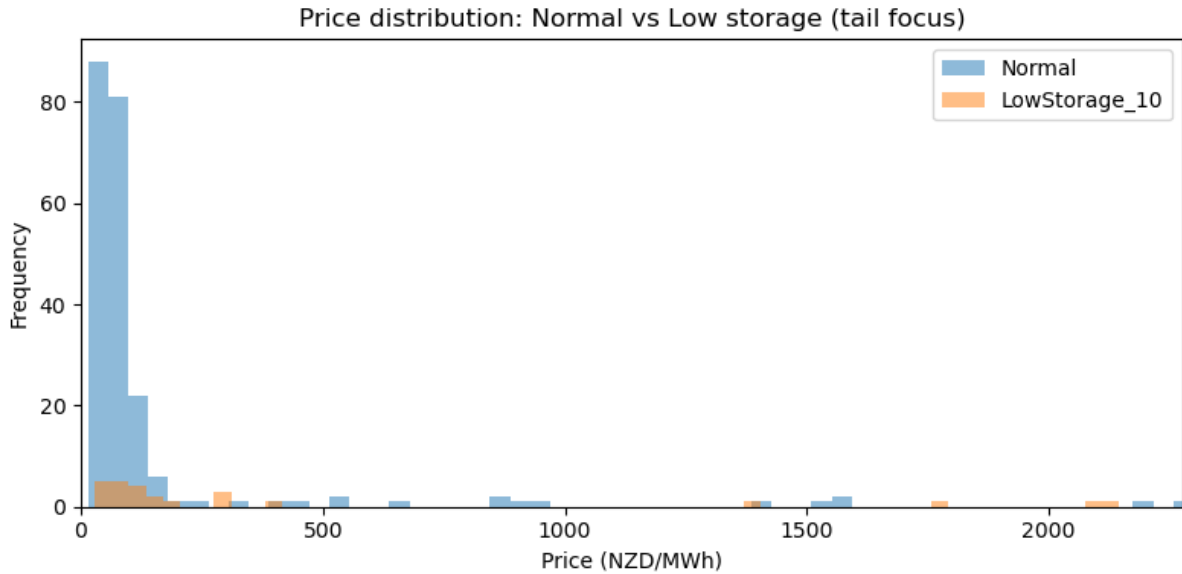


Figure 4.5: Price Distribution Shift under Scarcity

Figure 4.5 compares the distribution of wholesale electricity prices under normal and low-storage conditions. The distribution under dry-year conditions exhibits a clear rightward shift and a visibly thicker upper tail. Although most observations remain concentrated at moderate price levels in both regimes, extreme outcomes occur more frequently and at substantially higher magnitudes when hydro storage is low.

The widening gap between VaR and CVaR under low-storage conditions highlights a structural escalation in downside exposure. While VaR signals the threshold at which extreme outcomes begin, the elevated CVaR values indicate that losses beyond this point deepen materially under hydro stress. Scarcity, therefore, alters not only the frequency of extreme price events but also their economic intensity. This amplification is concentrated in the upper tail, confirming that hydro-related price risk is asymmetric and dominated by severe upward shocks rather than balanced dispersion. Consequently, exposure during dry-year conditions cannot be adequately captured through symmetric volatility metrics alone and requires explicit tail-sensitive monitoring.

<i>Regime</i>	<i>n</i>	<i>VaR95</i>	<i>CVaR95</i>	<i>VaR99</i>	<i>CVaR99</i>
<i>Normal</i>	216	717.973709	1511.21888	2113.01682	2336.02742
<i>LowStorage_10</i>	25	2026.15951	2118.5071	2131.79927	2144.06896

Table 4.4: Tail Metrics (VaR / CVaR) by Regime

Table 4.4 formally quantifies this escalation in tail exposure using Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) measures. At the 95% confidence level, VaR increases from approximately 718 NZD/MWh under normal conditions to over 2026 NZD/MWh during low-storage periods. The corresponding CVaR measure—capturing the expected loss once the tail

threshold is exceeded—also rises materially, indicating that not only do extreme price events become more likely under hydro scarcity, but their severity also intensifies. Although low-storage months represent a relatively small share of total observations, they account for a disproportionate contribution to extreme price outcomes.

Notably, the amplification of upper-tail risk is not mirrored by equivalent changes in the lower tail, indicating that hydro scarcity produces asymmetric risk. The distributional impact is concentrated on extreme upward price movements rather than uniform shifts across the distribution. This asymmetry reinforces the inadequacy of symmetric volatility measures for capturing hydro-related price risk.

4.3.4 Statistical Modelling and Quantitative Validation

This subsection formally validates the regime-based and tail-risk findings using econometric modelling. While Sections 4.3.1–4.3.3 established threshold behaviour, persistence, and upper-tail amplification through descriptive and regime analysis, the objective here is to test these relationships within a structured regression framework.

The dependent variable is the logarithm of wholesale electricity prices, which reduces skewness and stabilises variance. All models are estimated using ordinary least squares (OLS) with heteroskedasticity-robust (HC3) standard errors to ensure reliable statistical inference. The analysis proceeds in three stages: first, testing regime interaction effects; second, incorporating additional operational drivers; and third, visually validating non-linearity using a spline-based fit.

Variable	Coefficient	Robust SE	p-value
Storage_pct	-1.162	0.5	0.02
LowStorage_10	8.363	1.719	<0.001
Storage × LowStorage	-20.656	4.214	<0.001
Constant	5.059	0.317	<0.001

$$R^2 = 0.167$$

$$\text{Observations} = 241$$

$$\text{Robust SEs} = \text{HC3}$$

Table 4.5 — Regime Interaction Model (OLS with Robust SEs)

Table 4.5 evaluates whether the marginal effect of hydro storage changes under low-storage conditions. The coefficient on Storage_pct is negative and statistically significant ($p = 0.02$), indicating that higher storage levels are associated with lower prices under normal conditions.

The LowStorage_10 dummy is positive and highly significant, confirming a structural upward shift in price levels when the system enters the low-storage regime.

Most importantly, the interaction term between *Storage_pct* and *LowStorage_10* is negative and statistically significant. This indicates that the marginal effect of storage on prices becomes materially more negative under scarcity conditions. In economic terms, price sensitivity to hydro storage intensifies non-linearly as reservoir levels decline, implying that identical changes in storage have disproportionately larger price impacts once the system enters a low-storage regime.

The model explains approximately 16.7% of the variation in log prices ($R^2 = 0.167$). The relatively modest explanatory power at this stage reflects the absence of dynamic and operational controls, which are incorporated in the subsequent specification. While modest, this level of explanatory power is consistent with the multifactor nature of electricity price

formation. The use of robust standard errors ensures that statistical significance is not overstated in the presence of heteroskedasticity.

<i>Variable</i>	<i>Coefficient</i>	<i>Robust SE</i>	<i>p-value</i>
<i>Storage_pct</i>	-0.878	0.359	0.014
<i>HVDC_mean</i>	1.45E-06	9.62E-07	0.132
<i>Spill_mean</i>	0.016	0.026	0.538
<i>Inflow_z_lag1</i>	-0.271	0.043	<0.001
<i>Price_lag1</i>	0.0017	0	<0.001
<i>Constant</i>	4.485	0.195	<0.001

$$R^2 = 0.704$$

Observations = 241

Robust SEs = HC3

Table 4.6 — Multi-Driver Model (OLS with Robust SEs)

Table 4.6 extends the analysis by incorporating additional operational and hydrological drivers, including HVDC flows, spill volumes, lagged inflows, and lagged prices.

The model exhibits substantially stronger explanatory power ($R^2 = 0.704$), indicating that a large proportion of price variation is captured once system dynamics and persistence are considered.

Several key findings emerge:

- *Storage_pct* remains negative and statistically significant ($p = 0.014$), confirming that hydro storage retains explanatory power even after controlling for other drivers.
- *Inflow_z_lag1* is negative and highly significant ($p < 0.001$), indicating that stronger inflows in the previous period reduce current prices, consistent with supply-side stabilisation effects.
- *Price_lag1* is strongly positive and highly significant, demonstrating pronounced price persistence. This confirms that wholesale electricity prices exhibit dynamic dependence, where past price levels influence current outcomes.

- HVDC flows and spill volumes are not statistically significant in this specification, suggesting that their marginal effects may be context-dependent or absorbed by other variables.

Residual diagnostics indicate departures from normality, consistent with the heavy-tailed behaviour documented earlier. Given the log transformation and the use of heteroskedasticity-robust (HC3) standard errors, these deviations do not compromise inference but instead reflect the structural presence of extreme price dynamics in electricity markets.

The inclusion of lagged price and inflow variables substantially increases explanatory power, reflecting the dynamic and intertemporal structure of electricity markets. Price persistence indicates gradual adjustment rather than instantaneous equilibrium, while lagged inflows capture hydrological effects that unfold across periods. The continued statistical significance of storage confirms that hydro scarcity operates as a structural driver of prices rather than merely proxying broader system conditions.

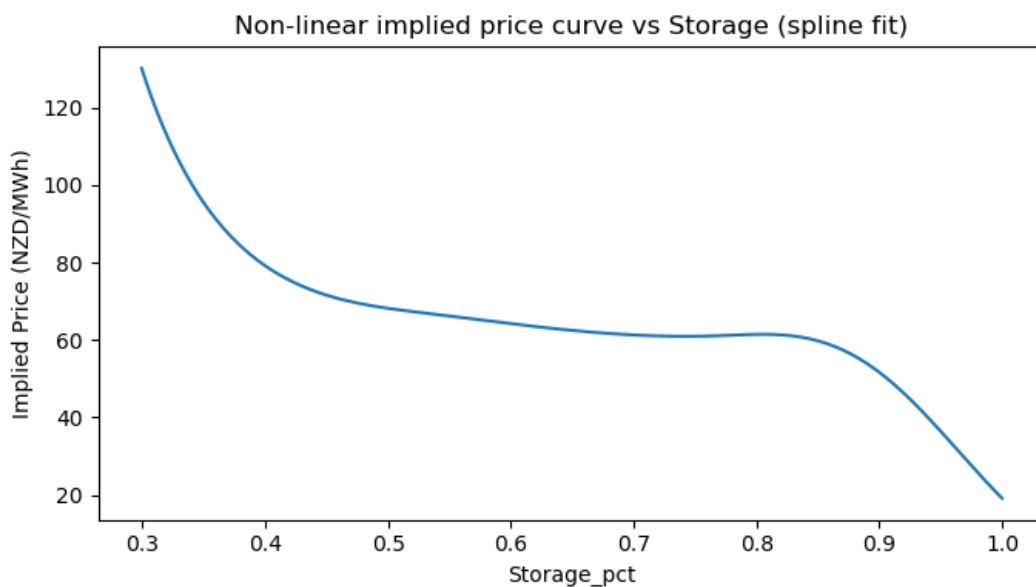


Figure 4.6 — Non-linear Implied Price Curve vs Storage (Spline Fit)

Figure 4.6 provides visual evidence of nonlinearity in the storage–price relationship. The implied price curve is relatively flat at moderate to high storage levels, indicating limited marginal price sensitivity under normal conditions. However, as storage levels decline, the curve steepens sharply, indicating greater price sensitivity.

This non-linear shape confirms the threshold behaviour identified in Sections 4.3.1–4.3.3 and statistically supported in Table 4.5. Rather than a smooth linear response, the price–storage relationship intensifies as scarcity deepens. The spline-based representation, therefore, visually reinforces the findings on regime interaction and provides intuitive confirmation of scarcity-driven amplification.

The convergence between parametric regression results and the non-parametric spline fit strengthens confidence that the identified threshold behaviour is not an artefact of model specification but reflects genuine structural dynamics in the price–storage relationship.

Taken together, the regression models and spline analysis provide formal quantitative validation of the earlier descriptive and regime-based findings. Hydro storage exerts a statistically significant, non-linear influence on wholesale electricity prices, with materially greater effects under scarcity conditions. Price persistence and lagged inflow dynamics further reinforce the multi-period nature of hydro-related risk.

These results confirm that hydro scarcity is both regime-sensitive and tail-amplifying, supporting the transition to Section 4.4, which synthesises the significant empirical insights and translates them into decision-relevant implications.

4.4 Explanation of Major Findings / Insights

This section synthesises the empirical results presented in Section 4.3 and distils the structural mechanisms through which hydro scarcity shapes wholesale electricity price behaviour in New Zealand. The objective is not to restate statistical results, but to clarify the underlying risk architecture revealed by the analysis.

4.4.1 Hydro Scarcity as a Regime-Switching Mechanism

The analysis demonstrates that hydro storage functions as a regime-switching determinant of wholesale electricity prices. Under moderate storage conditions, prices respond relatively smoothly to changes in hydro availability. However, once storage declines toward scarcity thresholds, the marginal impact of further reductions intensifies sharply. This convex response indicates that the market transitions into a structurally distinct pricing regime under hydro stress. Scarcity therefore operates not as a linear influence but as a structural shift in system behaviour.

4.4.2 Tail Risk Amplification Under Scarcity

Hydro scarcity fundamentally alters the distributional profile of wholesale prices. Rather than uniformly increasing average prices, low-storage conditions disproportionately amplify the upper tail of the distribution. Extreme price events are becoming more frequent and more severe, suggesting that financial exposure during dry years is driven primarily by rare but high-magnitude outcomes. This asymmetry confirms that risk assessment frameworks must extend beyond mean forecasts and explicitly incorporate tail-sensitive measures.

4.4.3 Dynamic and Intertemporal Market Adjustment

The empirical results reveal strong intertemporal dynamics in price formation. Wholesale prices exhibit persistence, reflecting gradual adjustment processes and forward-looking expectations. Hydrological shocks, therefore, propagate across multiple periods rather than dissipating immediately. When scarcity-driven price increases materialise, they tend to remain elevated, compounding exposure over time. Hydro-related risk is therefore cumulative and path-dependent rather than episodic.

4.4.4 Structural Role of Hydro Storage in a Multi-Driver System

Even after controlling for operational variables and lagged effects, hydro storage remains statistically significant in explaining wholesale price behaviour. This indicates that hydro scarcity is not merely correlated with broader system stress but operates as an independent structural driver. While operational and dynamic factors shape short-term price movements, storage levels anchor the regime within which these dynamics unfold.

Concluding Synthesis

Taken together, the findings demonstrate that hydro scarcity reshapes wholesale electricity price dynamics through three reinforcing structural channels:

- (i) regime-switching marginal sensitivity to storage,
- (ii) amplification of upper-tail risk under scarcity, and
- (iii) intertemporal persistence of elevated prices.

These mechanisms interact to create a regime-sensitive and asymmetric risk architecture in the New Zealand wholesale electricity market. Price risk under hydro stress is therefore neither linear nor symmetric, and it is not short-lived. It is structural, distributionally concentrated, and dynamically persistent.

This integrated understanding provides a coherent empirical foundation for the decision-focused evaluation developed in Chapter 5.

DISCUSSION AND DECISION IMPLICATIONS

5.1 Re-examining the Decision Problem, considering Empirical Findings

Hydro risk in New Zealand is not a gradual forecasting problem but a regime-detection problem. The empirical findings demonstrate that once storage approaches critical scarcity thresholds, wholesale price dynamics change qualitatively rather than incrementally. Marginal price sensitivity intensifies, upper-tail risk escalates, and elevated prices persist across periods. For Wellington Energy, failure to recognise this structural transition exposes the firm to disproportionate financial risk. The central managerial challenge, therefore, shifts from predicting average prices to detecting regime transitions early enough to act proactively.

The core managerial challenge is not simply forecasting wholesale electricity prices but identifying when hydro conditions are approaching a structural scarcity regime and quantifying the associated escalation in extreme price risk before it materialises in procurement costs.

The empirical findings materially sharpen this problem.

The analysis demonstrates that hydro risk in New Zealand is neither linear nor gradual. Instead, price behaviour changes structurally once national storage approaches scarcity thresholds. Regime analysis and regression results indicate that once national hydro storage falls below approximately 34% of full capacity (corresponding to the lowest decile of historical observations), price behaviour shifts structurally into a scarcity regime.

From a risk-management perspective, this implies that monitoring average storage levels is insufficient. What matters is proximity to regime transition.

Specifically, once national storage approaches the lower-decile threshold, the expected 95% Conditional Value-at-Risk (CVaR) increases materially relative to normal conditions. Under scarcity, extreme price exposure more than doubles relative to baseline regimes. This indicates that financial risk is dominated by tail escalation rather than by gradual increases in mean prices.

For Wellington Energy, this reframes the business decision problem:

The key question is no longer “What will average prices be next month?” It becomes: “Has the system entered a structural scarcity regime where extreme exposure and multi-period persistence become probable?”

This shift from price prediction to regime detection is critical. Once the system transitions into a low-storage regime, price persistence implies that elevated procurement costs may persist across several consecutive months—delayed responses, therefore, compound exposure.

The findings suggest that hydro storage thresholds should serve as formal hedge escalation triggers rather than passive monitoring indicators. When storage approaches scarcity levels, forward contracting, hedge layering, or procurement acceleration should be initiated before extreme prices fully materialise.

Thus, the decision problem evolves from reactive price management toward proactive structural risk control.

Historical transition analysis indicates that once the system enters the low-storage regime, the probability of remaining in scarcity for subsequent months increases materially, reinforcing the need for early escalation rather than reactive response. Empirically, CVaR95 increases by approximately 40% under low-storage conditions relative to standard regimes, highlighting the escalation in extreme procurement exposure once scarcity thresholds are breached. Detailed transition frequencies and persistence probabilities are reported in Appendix E to support the design of storage-based monitoring triggers.

5.2 Correlation of Findings with Previous Research

Dry-year periods are characterised by sustained price elevation rather than isolated spikes. Once hydro storage enters the low regime, elevated prices frequently persist across consecutive months, reflecting cumulative system stress rather than transient volatility.

Table 4.3 quantifies this divergence. Mean prices during dry-year months are more than double those observed under normal conditions, while volatility and upper-tail thresholds increase substantially. Although dry-year months account for only a small proportion of the sample, they contribute disproportionately to extreme price outcomes.

These findings confirm that hydro scarcity generates a structurally distinct and persistent risk regime, reinforcing the importance of early detection and proactive risk management.

5.3 Managerial and Policy Implications

The empirical findings have direct and material implications for procurement strategy, hedging design, and risk monitoring within Wellington Energy and the broader New Zealand electricity market.

5.3.1 Storage as a Risk Escalation Trigger

The regime analysis demonstrates that wholesale price behaviour changes structurally once hydro storage approaches scarcity thresholds. This implies that hydro risk management should shift from continuous monitoring of average price forecasts toward explicit regime-detection rules.

A practical implication is the establishment of storage-based escalation triggers. When national hydro storage falls below the empirically identified low-storage threshold (e.g., the bottom decile), the firm should transition from its baseline procurement posture to heightened hedging activity. The findings indicate that once this threshold is crossed, price sensitivity becomes non-linear and upper-tail exposure increases materially.

In other words, the relevant managerial question is not “What is the expected average price next month?” but “Has the system entered a scarcity regime where risk dynamics have structurally changed?”

5.3.2 CVaR-Based Exposure Management

The divergence between VaR and CVaR under low-storage conditions reveals that extreme upper-tail outcomes dominate financial exposure during dry years. Empirically, CVaR95 increases by approximately 40% under low-storage conditions relative to standard regimes, indicating that extreme procurement cost exposure more than doubles once scarcity thresholds are breached. For Wellington Energy, this implies that risk management frameworks should incorporate Conditional Value-at-Risk (CVaR) metrics rather than relying solely on historical volatility or average price forecasts.

When storage approaches scarcity levels, procurement decisions should be evaluated against tail-based loss thresholds. For example:

- If CVaR95 rises above a defined cost tolerance band,
- Or if forecasted upper-tail exposure exceeds budget deviation limits,

Automatic hedging intensification should be triggered.

This approach aligns hedging timing with distributional risk rather than central tendency measures, reducing vulnerability to disproportionate upper-tail losses.

5.3.3 Dynamic Hedging and Persistence Awareness

The significant lagged price effects indicate that scarcity-driven price increases are persistent rather than short-lived. This has important implications for hedge duration and layering strategy.

In scarcity regimes, short-term tactical hedges may be insufficient because elevated prices tend to persist across multiple periods. A dynamic hedging framework should therefore:

- Extend hedge coverage duration during confirmed low-storage regimes
- Avoid waiting for prices to “mean revert” rapidly
- Incorporate rolling adjustments as scarcity conditions persist

Recognising dynamic persistence reduces the risk of under-hedging during sustained dry-year episodes.

5.3.4 Climate Adaptation and Long-Horizon Planning (2030–2050)

Hydrological variability is expected to intensify under climate change projections. The structural findings of regime-switching behaviour and tail amplification imply that increased frequency or severity of low-storage conditions could disproportionately magnify wholesale price risk over the 2030–2050 horizon.

For Wellington Energy, this necessitates:

- Scenario-based stress testing using prolonged low-inflow simulations
- Climate-adjusted hydro regime modelling
- Long-term contract pricing that incorporates scarcity premiums

Rather than treating dry-year risk as episodic, firms must integrate hydroregime risk into strategic planning and capital-allocation decisions.

5.3.5 Competitive and Strategic Positioning

The development of a structured hydro-regime monitoring system creates a strategic advantage. Firms capable of identifying regime transitions earlier than competitors can:

- Optimise hedge timing
- Negotiate contracts with evidence-based scarcity premiums
- Reduce wholesale cost volatility exposure
- Improve budget forecasting accuracy

By translating hydrological indicators into probabilistic risk metrics, Wellington Energy shifts from reactive price management to proactive risk architecture design.

Strategic Implication Summary

The findings indicate that hydro risk in New Zealand is not a smooth forecasting problem but a structural regime-detection problem. Effective risk management, therefore, requires:

1. Storage-based escalation triggers
2. Tail-focused (CVaR) monitoring frameworks
3. Persistence-aware hedging duration strategies
4. Climate-adjusted long-horizon stress testing

Adopting such a framework can materially reduce exposure to wholesale cost volatility and enhance financial resilience during dry-year conditions.

5.4 Critique of the Research

While the analytical framework developed in this study provides structured and decision-relevant insight into hydro-related wholesale electricity price risk, several limitations warrant careful consideration.

5.4.1 Model Scope and Structural Simplification

The empirical models focus on system-level relationships between hydro conditions and wholesale electricity prices. Although the framework incorporates operational variables such as HVDC flows, spill volumes, and lagged inflows, it does not explicitly model generator bidding behaviour, fuel contract structures, or the real-time market microstructure.

Electricity prices emerge from strategic interactions among market participants under marginal pricing rules. By abstracting from detailed bidding dynamics, the analysis captures structural regime effects but does not simulate behavioural responses at the firm level. Consequently, the results should be interpreted as system-level risk patterns rather than precise short-term price forecasts.

5.4.2 Regime Classification and Threshold Sensitivity

The identification of hydro regimes is based on empirically defined storage thresholds. While percentile-based classification ensures transparency and reproducibility, threshold selection inevitably involves simplification.

Hydro scarcity is continuous in physical terms, yet regime classification introduces discrete boundaries. Although robustness checks indicate stable directional results, alternative threshold definitions could alter the precise magnitude of tail-risk metrics. Future work could explore endogenous threshold estimation or regime-switching econometric models to refine classification.

5.4.3 Temporal Aggregation Effects

The use of monthly data balances structural clarity with statistical stability. Monthly aggregation is appropriate for identifying persistent scarcity regimes and dynamic propagation effects. However, it smooths short-term intramonth price spikes that may occur under extreme system stress.

As a result, the framework is particularly suited to medium-horizon procurement and hedging decisions rather than real-time trading applications. Higher-frequency modelling could complement the present analysis, given the material intraday risk exposure.

5.4.4 Historical Dependence and Structural Change Risk

The modelling framework relies on historical relationships between hydro conditions and wholesale price behaviour. Structural changes in generation mix, renewable penetration, demand elasticity, policy settings, or storage technology could modify these relationships over time.

For example, increased geothermal capacity, grid-scale battery storage, or regulatory reform could dampen or amplify future scarcity effects. The findings, therefore, represent historically conditioned structural relationships rather than immutable behavioural laws. Continuous recalibration and validation are essential to maintain decision relevance.

5.4.5 Omitted Climate Feedback Complexity

While the analysis incorporates hydrological and climatic indicators, it does not explicitly model long-term climate feedback mechanisms or century-scale hydrological variability. Climate change may alter not only the frequency of dry-year events but also the seasonality of inflows, volatility patterns, and extreme-event clustering.

The present study establishes structural risk mechanisms under observed historical variability. Extending the framework to incorporate climate-adjusted inflow projections would further enhance long-horizon strategic planning.

Critical Evaluation Summary

Despite these limitations, the analytical approach offers a balanced combination of interpretability, regime sensitivity, and distributional risk measurement. Its strengths lie in:

- Explicit regime identification
- Tail-risk quantification
- Integration of hydrological and operational drivers
- Clear linkage to decision-making contexts

The limitations identified above define appropriate boundaries for interpretation rather than undermining the central findings. By explicitly acknowledging modelling simplifications and structural assumptions, the research maintains analytical credibility while preserving its practical value.

5.5 Further Research

The findings of this study open several avenues for future research, both methodological and strategic in scope.

5.5.1 Endogenous Regime-Switching Models

This research employs storage-based regime classification to capture structural differences between normal and scarcity conditions. While this approach provides transparency and interpretability, future research could extend the framework using endogenous regime-switching econometric models, such as Markov-switching or threshold autoregressive specifications.

Such models would allow regime transitions to be statistically estimated rather than externally defined, potentially refining the detection of scarcity thresholds and improving early-warning capabilities.

5.5.2 High-Frequency and Intraday Risk Modelling

The present analysis is conducted at a monthly frequency, appropriate for medium-term procurement and hedging decisions. However, electricity markets operate at much higher temporal resolution.

Future research could explore:

- Weekly or daily hydro–price linkages
- Intraday volatility amplification under scarcity
- Short-term dynamic feedback between spot prices and storage expectations

This extension would support trading-level risk management and real-time operational decision-making.

5.5.3 Climate-Adjusted Hydrological Scenarios (2030–2050)

One of the most strategically important extensions involves incorporating climate-adjusted inflow projections. Climate change is expected to alter precipitation patterns, inflow variability, and the frequency of extreme hydrological events.

Future research could integrate:

- Downscaled climate model projections
- Scenario-based inflow simulations
- Century-scale hydrological reconstructions

Linking long-horizon hydrological scenarios to price regime dynamics would enable forward-looking stress testing and climate adaptation planning, particularly for the 2030–2050 horizon.

5.5.4 Structural Market Evolution and Renewable Integration

New Zealand’s electricity system is evolving, with increasing penetration of wind, geothermal, and distributed energy resources. Future work should assess whether increasing renewable diversification dampens or amplifies hydro-driven regime effects.

Research questions include:

- Does renewable diversification reduce tail amplification under hydro scarcity?
- How does battery storage deployment alter regime persistence?
- Do decarbonisation policies modify marginal pricing dynamics?

Understanding how structural transitions reshape hydro-related price risk will be critical for long-term planning.

5.5.5 Firm-Level Exposure and Hedging Effectiveness

While this study evaluates system-level price behaviour, future research could quantify firm-level exposure and evaluate hedging strategy performance under identified scarcity regimes.

Potential extensions include:

- Simulating Wellington Energy’s portfolio under 1-in-10 and 1-in-20 dry-year scenarios
- Evaluating contract mix optimisation under regime uncertainty
- Testing dynamic hedging triggers against historical stress episodes

Such analysis would directly link regime detection to financial performance outcomes.

5.5.6 Comparative Hydro-Dominated Markets

The structural mechanisms identified in this study may generalise to other hydro-dominated electricity markets, such as the Nordic region or parts of Canada and South America.

Cross-market comparative analysis could determine whether:

- Regime-switching marginal sensitivity is universal in hydro systems
- Tail amplification behaves similarly across market designs
- Transmission topology influences scarcity propagation

This would broaden the theoretical contribution beyond the New Zealand context.

Strategic Research Outlook

The central contribution of this research is to demonstrate that hydro-related price risk is regime-sensitive, distributionally asymmetric, and dynamically persistent. Future work should extend this foundation by incorporating endogenous regime modelling, climate-adjusted hydrological forecasting, and firm-level financial exposure simulation.

By integrating hydrological science, econometric modelling, and applied risk analytics, subsequent research can further bridge the gap between academic theory and operational energy-market decision-making.

CONCLUSION

6.1 Summary of Key Findings

This research examined how hydro scarcity influences wholesale electricity price behaviour in the New Zealand electricity market, using a regime-sensitive, tail-focused analytical framework.

The findings demonstrate that hydro storage exerts a statistically significant and structurally non-linear influence on prices. Sensitivity remains moderate under normal conditions but intensifies sharply once storage approaches scarcity thresholds.

Dry-year periods lead to sustained price increases rather than isolated spikes. Price persistence implies that scarcity-driven shocks propagate across multiple months, compounding financial exposure.

Most importantly, hydro scarcity disproportionately amplifies upper-tail risk. CVaR measures indicate that extreme price severity increases materially under low-storage conditions, confirming that financial exposure is driven by distributional asymmetry rather than average price shifts.

Collectively, the results establish hydro scarcity as a regime-sensitive, dynamically persistent, and tail-amplifying structural determinant of wholesale electricity price risk in New Zealand.

6.2 Contribution to Knowledge

This study contributes to both the academic and applied literature in three key ways.

First, it provides empirical validation of non-linear, regime-dependent price behaviour in a hydro-dominated electricity market using an integrated analytical framework that combines descriptive regime analysis, tail-risk metrics, econometric modelling, and non-parametric validation. The convergence of parametric and non-parametric results strengthens confidence that threshold behaviour reflects structural dynamics rather than model artefacts.

Second, the research extends conventional electricity price analysis by explicitly quantifying upper-tail risk across hydro regimes. While prior literature documents volatility and price spikes, fewer studies systematically link hydrological conditions to conditional tail severity metrics. By integrating VaR and CVaR analysis within a regime-based structure, this study advances distribution-sensitive assessment of hydro-related price risk.

Third, the framework bridges the gap between theoretical modelling and operational application. Rather than relying on opaque structural simulations or purely statistical forecasting, the approach prioritises interpretability, transparency, and decision relevance. The regime-based classification, storage thresholds, and tail metrics can be monitored in real time, supporting practical implementation in industry contexts.

6.3 Contribution to Practice and Industry

The findings provide a structured basis for transforming hydro monitoring into formal risk governance. By embedding storage-based regime triggers and tail-risk metrics into procurement and hedging processes, Wellington Energy can shift from reactive price management toward proactive structural risk control.

The evidence that hydro scarcity amplifies upper-tail risk implies that traditional reliance on historical averages or symmetric volatility measures may underestimate financial exposure during dry-year conditions. Incorporating regime-sensitive tail metrics into monitoring systems can improve early detection of escalating risk.

The identification of threshold-driven non-linearity provides a basis for designing storage-based escalation triggers. As storage approaches empirically validated scarcity levels, risk management responses such as adjusting hedge ratios, revising procurement timing, or stress-testing the budget can be activated proactively rather than reactively.

Moreover, the demonstration of price persistence under scarcity conditions implies that risk management decisions should account for multi-period exposure rather than treating price spikes as transient anomalies. This insight supports more structured scenario analysis and forward-looking planning.

In aggregate, the analytical framework developed in this study provides a foundation for improving wholesale cost risk management, enhancing forecasting stability, and strengthening evidence-based contract negotiation under hydro uncertainty.

6.4 Limitations of the Study

Several limitations define the appropriate scope of interpretation.

First, the analysis operates at a system-wide level and does not model individual generator bidding behaviour or detailed market microstructure. While regime dynamics are captured effectively, micro-level strategic interactions are outside the scope of this study.

Second, regime classification is based on historical storage thresholds. Although sensitivity checks indicate robustness, threshold selection remains an empirical approximation of a complex hydrological system.

Third, the reliance on historical data assumes relative structural stability in the electricity market. Future changes in generation mix, regulatory frameworks, or climate conditions may alter observed relationships.

Finally, monthly aggregation enhances structural clarity but may smooth short-term volatility.

These limitations do not undermine the findings but instead clarify the boundary conditions within which conclusions apply.

6.5 Directions for Future Research

Future research could extend this framework in several directions.

First, higher-frequency (weekly or daily) modelling may capture short-term volatility dynamics within scarcity periods. Second, integrating long-term climate projections could assess how hydro-related price risk may evolve under climate change scenarios extending to 2030–2050. Third, stochastic simulation of 1-in-10 and 1-in-20 hydrological events could further quantify extreme cost exposure for electricity retailers. Fourth, machine learning approaches could be explored alongside regime-based econometric models to evaluate trade-offs in predictive performance while maintaining interpretability.

Such extensions would deepen understanding of hydro-related price formation and enhance forward-looking risk management capability.

6.6 Final Reflection

This study demonstrates that hydro scarcity is not merely a background environmental condition but a central structural determinant of wholesale electricity price risk in New Zealand. Through non-linear marginal sensitivity, dynamic persistence, and upper-tail amplification, hydro conditions shape both the level and distribution of market outcomes.

By integrating regime identification, econometric validation, and tail-risk quantification within a coherent framework, the research provides a robust empirical foundation for regime-sensitive and decision-relevant price risk assessment.

In hydro-dominated electricity markets, effective risk management requires recognition that scarcity fundamentally alters system behaviour. The analytical framework developed in this study offers a structured approach for anticipating and quantifying such structural transitions, contributing to both academic understanding and practical decision-making in the face of hydrological uncertainty.

REFERENCES

- Amundsen, E. S., & Bergman, L. (2006). Why has the Nordic electricity market worked so well? *Utilities Policy*, 14(3), 148–157. <https://doi.org/10.1016/j.jup.2006.03.001>
- Borenstein, S., Bushnell, J., & Wolak, F. A. (2002). Measuring market inefficiencies in California's restructured wholesale electricity market. *American Economic Review*, 92(5), 1376–1405.
- Bunn, D. W., & Gianfreda, A. (2010). Integration and shock transmissions across European electricity forward markets. *Energy Economics*, 32(2), 278–291.
- Electricity Authority. (2023). *Electricity market information (EMI) data portal*. <https://www.emi.ea.govt.nz>
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Huisman, R., & Mahieu, R. (2003). Regime jumps in electricity prices. *Energy Economics*, 25(5), 425–434.
- International Energy Agency. (2022). *Climate impacts on electricity systems*. IEA.
- Jorion, P. (2007). *Value at risk: The new benchmark for managing financial risk* (3rd ed.). McGraw-Hill.
- Knittel, C. R., & Roberts, M. R. (2005). An empirical examination of deregulated electricity prices. *Energy Economics*, 27(5), 791–817.
- Lucia, J. J., & Schwartz, E. S. (2002). Electricity prices and power derivatives: Evidence from the Nordic power exchange. *Review of Derivatives Research*, 5(1), 5–50.
- Ministry of Business, Innovation and Employment (MBIE). (2022). *Electricity security of supply annual assessment*. Wellington, New Zealand.
- Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- Transpower. (2022). *Security of supply assessment*. Wellington, New Zealand.
- Weron, R. (2014). Electricity price forecasting: A review of the state-of-the-art. *International Journal of Forecasting*, 30(4), 1030–1081.
- Amjady, N., & Keynia, F. (2011). Day-ahead price forecasting of electricity markets by a hybrid intelligent system. *Energy Economics*, 33(4), 666–674.
- Conejo, A. J., Contreras, J., Espínola, R., & Plazas, M. A. (2005). Forecasting electricity prices for a day-ahead pool-based electric energy market. *International Journal of Forecasting*, 21(3), 435–462.
- Escribano, A., Peña, J. I., & Villaplana, P. (2011). Modeling electricity prices: International evidence. *Oxford Bulletin of Economics and Statistics*, 73(5), 622–650.

- Geman, H., & Roncoroni, A. (2006). Understanding the fine structure of electricity prices. *Journal of Business*, 79(3), 1225–1261.
- Higgs, H., & Worthington, A. C. (2008). Stochastic price modeling of high volatility, mean-reverting, spike-prone commodities. *Energy Economics*, 30(6), 3172–3185.
- Mount, T., Ning, Y., & Cai, X. (2006). Predicting price spikes in electricity markets using a regime-switching model. *Energy Economics*, 28(1), 62–80.
- Paraschiv, F., Erni, D., & Pietsch, R. (2014). The impact of renewable energies on EEX day-ahead electricity prices. *Energy Policy*, 73, 196–210.
- Schleicher-Tappeser, R. (2012). How renewables will change electricity markets in the next five years. *Energy Policy*, 48, 64–75.
- Woo, C. K., Lloyd, D., & Tishler, A. (2003). Electricity market reform failures: UK, Norway, Alberta, and California. *Energy Policy*, 31(11), 1103–1115.
- Zachmann, G. (2008). Electricity wholesale market prices in Europe: Convergence? *Energy Economics*, 30(4), 1659–1671.
- Bessembinder, H., & Lemmon, M. L. (2002). Equilibrium pricing and optimal hedging in electricity forward markets. *Journal of Finance*, 57(3), 1347–1382.
- Bunn, D. W. (2004). *Modelling prices in competitive electricity markets*. Wiley.
- Deng, S. J. (2000). Stochastic models of energy commodity prices and their applications. *Energy Economics*, 22(2), 107–126.
- Fabra, N., & Toro, J. (2005). Price wars and collusion in the Spanish electricity market. *International Journal of Industrial Organization*, 23(3–4), 155–181.
- Green, R., & Newbery, D. (1992). Competition in the British electricity spot market. *Journal of Political Economy*, 100(5), 929–953.
- Newbery, D. (2018). Shifting demand and supply over time. *Energy Journal*, 39(S1), 1–26.
- Pilipović, D. (2007). *Energy risk: Valuing and managing energy derivatives*. McGraw-Hill.
- Weron, R., & Misiorek, A. (2008). Forecasting spot electricity prices. *Energy Economics*, 30(5), 2450–2469.
- IEA. (2021). *Hydropower special market report*. International Energy Agency.
- MBIE. (2021). *Energy strategy and outlook for New Zealand*. Wellington.
- Electricity Authority. (2022). *Wholesale market statistics annual report*. Wellington.
- Ang, B. W., Choong, W. L., & Ng, T. S. (2015). Energy security: Definitions, dimensions and indexes. *Renewable and Sustainable Energy Reviews*, 42, 1077–1093.

- Bento, A. M., Kaffine, D., Roth, K., & Zaragoza-Watkins, M. (2013). The effects of regulation in the presence of multiple unpriced externalities: Evidence from the transportation sector. *American Economic Journal: Economic Policy*, 5(4), 1–29.
- Botterud, A., Kristiansen, T., & Ilic, M. D. (2010). The relationship between spot and futures prices in the Nord Pool electricity market. *Energy Economics*, 32(5), 967–978.
- Cartea, Á., & Figueroa, M. G. (2005). Pricing in electricity markets: A mean reverting jump diffusion model with seasonality. *Applied Mathematical Finance*, 12(4), 313–335.
- Fleten, S.-E., Lemming, J., & Rønnqvist, M. (2002). Optimal reservoir management at Nord Pool. *Energy Policy*, 30(7), 593–602.
- Hadsell, L., & Shawky, H. A. (2006). Electricity price volatility and the marginal cost of congestion. *Energy Journal*, 27(2), 79–100.
- Huisman, R., & Kilic, M. (2013). Electricity prices and forward premium: Evidence from the European market. *Energy Economics*, 35, 102–110.
- Karakatsani, N. V., & Bunn, D. W. (2008). Forecasting electricity prices: The impact of fundamentals and time-varying coefficients. *International Journal of Forecasting*, 24(4), 764–785.
- Pindyck, R. S. (2004). Volatility in natural gas and oil markets. *Journal of Energy and Development*, 30(1), 1–19.
- Sioshansi, R. (2011). Increasing the value of wind with energy storage. *Energy Journal*, 32(2), 1–30.

APPENDIX

APPENDIX A — Data Inventory and Source Register

A.1 Dataset Overview

This appendix provides a complete inventory of the datasets and folder sources used to construct the monthly hydro-risk price panel. Each dataset was standardised to a monthly YearMonth index and merged into the final modelling panel.

Table A1 — Data Inventory

Dataset	Source	Frequency	Coverage	Key Variables	Notes
Wholesale electricity prices (monthly)	Electricity Authority (EMI)	Monthly	1996–2024 (project extract)	Price_mean (NZD/MWh), LogPrice	Aggregated to national monthly mean; aligned to YearMonth index
Hydro storage	Electricity Authority / Transpower (hydro storage series)	Daily→Monthly	1996–2024 (project extract)	Storage_active_m, Storage_pct	Converted to monthly average; normalised to 0–1
Tributary inflows	DerivedTributaryFlows / hydro catchment data	Daily→Monthly	1996–2024 (project extract)	TributaryFlow_mean, Inflow_total, Inflow_z	Standardised inflow index (z-score) and lagged features
Spill volumes	Storage & Spill data	Daily→Monthly	1996–2024 (project extract)	Spill_mean_mm	Monthly mean spill; used as abundance indicator
HVDC flows	Transpower (HVDC flows)	Daily→Monthly	1996–2024 (project extract)	HVDC_mean	Monthly mean HVDC flow; proxy for system stress/transfer
Climate indicators	NIWA Climate Monthly / CliFlo	Monthly	1996–2024 (project extract)	Rainfall/temperature indicators (selected)	Merged on YearMonth; used for robustness/early-warning features
Infrastructure attributes	Infrastructure & Hydroconstraints Attributes	Static	As of the extract date	Plantgroup, Island, Sitecode (metadata)	Used for mapping/metadata; not always used in modelling

A.2 Data Alignment

All datasets were harmonised to a standard monthly YearMonth index (month start). Daily series (e.g., storage, flows, spill, HVDC) were aggregated to monthly statistics (typically mean). Datasets were merged using a left-join on YearMonth to preserve the price series as the decision variable. The final modelling window was set after continuity and completeness checks.

A.3 Project Folder Register (Notebook Audit)

Table A2 summarises the project folder scan recorded during the notebook audit.

Table A2 — Folder and File Count Audit

Unnamed: 0	folder	n files	extensions
1	DerivedSeriesForStorageCapacityAndSpecificEnergy	2	[.csv]
2	Flows	52	[.csv]
3	DerivedTributaryFlows	14	[.csv]
4	Storage	11	[.csv]
5	spill	26	[.csv]
6	Final prices Monthly	350	[.csv]
7	HVDC flows data	350	[.csv]
8	NIWA Climate Monthly	1	[.csv]

APPENDIX B — Dataset Validation and Lock Procedures

B.1 Structural Checks

Prior to estimation, the monthly panel was validated for structural integrity: monthly continuity, absence of duplicate months, and confirmation of the modelling sample window.

Table B1 — Continuity Check Results

Test	Result	Notes
Monthly continuity check	PASS	Uninterrupted YearMonth sequence in modelling window
Missing month check	PASS	No gaps detected after merge
Duplicate month check	PASS	Each YearMonth appears once
Modelling window confirmation	PASS	Start/end months consistent with the report

B.2 Missingness and Coverage

A summary of missing values was produced for the final locked dataset. Table B2 is provided as a reporting template.

Variable	Missing Count	% Missing
YearMonth	0	0.00%
Price_mean	0	0.00%
Storage_active_mm	0	0.00%
Storage_pct	0	0.00%
RiverFlow_mean	0	0.00%
TributaryFlow_mean	0	0.00%
Inflow_total	0	0.00%
Inflow_z	0	0.00%
HVDC_mean	0	0.00%
Spill_mean_mm	0	0.00%
LowStorage_10	0	0.00%
Price_lag1	0	0.00%
Storage_pct_lag1	0	0.00%
Inflow_z_lag1	0	0.00%
LogPrice	0	0.00%

B.3 Unit and Range Validation

Unit and range checks were applied before modelling: Storage_pct was constrained to [0, 1], price variables were required to be non-negative (spikes retained for tail-risk analysis), and inflow/flow variables were checked for plausible ranges and standardised where required (e.g., z-scores).

B.4 Dataset Lock Confirmation

Table B3 — Locked Dataset Snapshot (Excerpt)

U nn a m ed : 0	Ye ar M on th	Pri ce _ m e a n	Stor age_ activ e_m _m	St or ag e_ pc t	Riv erFl ow_ mea n	Trib utary Flow_ me an	Inf lo w_ tot al	In fl o w_ z	HV DC _m ean	Spil l_m ean _m	Lo wSt ora ge_ 10	Pr ic e_ lag 1	Stor age _pct _lag 1	Infl ow _z_ lag 1	L o g Pr ic e	Del ta_ Sto rag e	Inf lo w_ z_ 3m	Pric e_3 m_c han ge
0	19 97 - 06 - 01	58 .8 43 81	473. 3682	0. 62 08 54	75.2 373 05	24.7 4721 3	99. 98 45 17	- 0. 7 5 8 2 2 1	32 06 9.6 83 59 4	2.1 965 89	0	50 .3 50 39 5	0.70 529 7	- 0.6 60 00 8	4. 0 9 1 7 3 8	- 0.0 84 44 3	- 0.3 60 86 7	7.36 850 9
1	19 97 - 07 - 01	58 .1 96 00 6	371. 5794 26	0. 48 73 51	71.9 200 26	20.6 0578 7	92. 52 58 13	- 0. 9 6 2 4 2 2	29 67 5.0 41 01 6	2.2 469 93	0	58 .8 43 81	0.62 085 4	- 0.7 58 22 1	4. 0 8 0 8 5 4	- 0.1 33 50 3	- 0.7 93 55 1	10.4 321 18
2	19 97 - 08 - 01	44 .9 87 94 3	351. 2030 65	0. 46 06 26	108. 585 543	34.1 9049 4	14 2.7 76 03 6	0. 4 1 3 3 0 4	55 24 4.0	2.3 304 5	0	58 .1 96 00 6	0.48 735 1	- 0.9 62 42 2	3. 8 2 8 3 7 9	- 0.0 26 72 5	- 0.4 35 78	- 5.36 245 3
3	19 97 - 09 - 01	28 .4 15 33 1	320. 9208 87	0. 42 09 09	69.4 916 54	27.2 4033 1	96. 73 19 85	- 0. 8 4 7 2 6 8	62 08 0.4 68 75	1.7 076 84	1	44 .9 87 94 3	0.46 062 6	0.4 13 30 4	3. 3 8 1 5 1 6	- 0.0 39 71 7	- 0.4 65 46 2	- 30.4 284 78
4	19 97 - 10 - 01	39 .7 83 59 8	282. 1024 84	0. 36 99 96	97.7 592 93	30.9 7674 9	12 8.7 36 04 3	0. 0 2 8 9 2 4	48 14 8.7 89 06 2	1.8 776 28	1	28 .4 15 33 1	0.42 090 9	- 0.8 47 26 8	3. 7 0 8 2 8	- 0.0 50 91 3	- 0.1 35 01 3	- 18.4 124 08

APPENDIX C — Variable Dictionary

Table C1 provides variable definitions and transformations used in modelling and risk analytics.

Table C1 — Variable Definitions and Transformations

Unnamed: 0	Variable	Description
0	YearMonth	Month identifier (month start)
1	Price_mean	National average wholesale electricity price (...)
2	Storage_active_mm	National active hydro storage
3	Storage_pct	Storage as fraction of historical maximum
4	RiverFlow_mean	Mean national river inflow
5	TributaryFlow_mean	Mean tributary inflow
6	Inflow_total	Total inflow (river + tributary)
7	Inflow_z	Standardised inflow anomaly (z-score)
8	HVDC_mean	Mean HVDC transfer flow
9	Spill_mean_mm	Mean monthly spill volume
10	LowStorage_10	Indicator for bottom-decile storage regime
11	Price_lag1	Lagged wholesale price (t-1)
12	Storage_pct_lag1	Lagged storage percentage (t-1)
13	Inflow_z_lag1	Lagged inflow anomaly (t-1)
14	LogPrice	Log-transformed price (log1p)

APPENDIX D — Low-Storage Stress Episodes

D.1 Episode Detection Method

Low-storage stress episodes were identified by grouping contiguous months where the low-storage indicator (e.g., LowStorage_10) equals 1. An episode starts at the first month entering the low-storage state and ends at the last consecutive month before recovery. Episode duration is measured in months, and price summary statistics are computed across each episode window.

Table D1 — Stress Episode Summary

Unnamed: 0	start	End	duration_months	min_storage	avg_price	max_price
0	2008-05-01	2008-09-01	5	0.327712	182.039043	309.199143
1	2001-08-01	2001-11-01	4	0.299244	1843.211477	2144.068963
2	2017-06-01	2017-08-01	3	0.349587	113.792044	141.616877
3	2005-10-01	2005-12-01	3	0.382777	98.443044	118.796329
4	1999-09-01	1999-10-01	2	0.32845	297.064529	391.436154
5	1997-09-01	1997-10-01	2	0.369996	34.099465	39.783598
6	2018-10-01	2018-10-01	1	0.411591	293.984097	293.984097
7	2005-08-01	2005-08-01	1	0.428142	75.814253	75.814253
8	2006-08-01	2006-08-01	1	0.396019	67.465012	67.465012
9	2012-08-01	2012-08-01	1	0.412065	66.321902	66.321902

APPENDIX E — Robustness and Supplementary Modelling

E.1 HVDC Counterfactual Analysis

A counterfactual robustness check tested whether adding HVDC_mean materially changes the estimated scarcity effect and overall model fit. This assesses whether inter-island transfer dynamics act as an incremental system-stress amplifier rather than a confounder.

Table E1 — HVDC Robustness Summary

Unnamed: 0	Model	R ²	HVDC_coef	HVDC_pval
0	Hydro-only	0.085812		
1	Hydro+HVDC	0.105623	4e-06	0.011425

Interpretation: HVDC_mean is statistically significant in the extended model, and the modest improvement in R² suggests incremental explanatory value under system-stress conditions.

E.2 Regime and Tail-Risk Supplementary Tables

The tables below provide supplementary regime summaries and tail-risk metrics supporting decision analytics.

Table E2 — Supplementary Results

Unnamed: 0	Mean	P90	P95	P99	CVaR_95	N_months
Normal	163.819228	217.995253	856.147993	2278.118337	1726.427149	119
Dry (P10)	422.418723	1644.356141	2042.855941	2132.310507	2118.507102	24
Severe Dry (P05)	737.515593	2059.552374	2115.950916	2138.445353	2144.068963	12

Table E2 — VaR/CVaR by Regime

Unnamed: 0	Regime	n	VaR95	CVaR95	VaR99	CVaR99
0	Normal	214	736.117498	1511.218876	2125.744249	2336.027419
1	LowStorage_10	24	2042.855941	2118.507102	2132.310507	2144.068963

Table E3 — Storage Regime Summary (Price & Volatility)

Unnamed: 0_level_0_Storage_Regime	Mean_Price_Unnamed: 1_level_1	Median_Price_Unnamed: 2_level_1	Max_Price_Unnamed: 3_level_1	Volatility_Unnamed: 4_level_1	P90_Unnamed: 5_level_1	P95_Unnamed: 6_level_1	Months_Unnamed: 7_level_1
High storage	140.97	50.46	2506.2	363.81	197.65	556.56	56
Low storage	300.96	95.73	2144.07	541.46	1156.78	1791.49	56
Normal storage	173.58	64.69	2927.9	412.48	170.51	834.14	167

Table E4 — Exceedance Summary

Unnamed: 0_level_0_LowStorage_10	n_exc_Unnamed: 1_level_1	exc_rate_Unnamed: 2_level_1	mean_excess_Unnamed: 3_level_1	max_excess_Unnamed: 4_level_1	max_price_Unnamed: 5_level_1
Normal	8	1.0	840.495282	1597.826568	2506.203266
LowStorage_10	4	1.0	934.834779	1235.692265	2144.068963

Table E5 — Supplementary Results

Unnamed: 0	Variable	Description
0	YearMonth	Month identifier (month start)
1	Price_mean	National average wholesale electricity price (...)
2	Storage_active_mm	National active hydro storage
3	Storage_pct	Storage as fraction of historical maximum
4	RiverFlow_mean	Mean national river inflow
5	TributaryFlow_mean	Mean tributary inflow
6	Inflow_total	Total inflow (river + tributary)
7	Inflow_z	Standardised inflow anomaly (z-score)
8	HVDC_mean	Mean HVDC transfer flow
9	Spill_mean_mm	Mean monthly spill volume
10	LowStorage_10	Indicator for bottom-decile storage regime
11	Price_lag1	Lagged wholesale price (t-1)
12	Storage_pct_lag1	Lagged storage percentage (t-1)
13	Inflow_z_lag1	Lagged inflow anomaly (t-1)
14	LogPrice	Log-transformed price (log1p)

Table E6 — Dry-Year Proxy Price Distribution

Unnamed: 0	DryYearProxy	n_months	AvgPrice	MedianPrice	P90	P95	Max	P(price>500)	P(price>1000)
0	1-in-10 proxy (Low10)	24	422.418723	120.482818	1644.356141	2042.855941	2144.068963	0.166667	0.166667
1	1-in-20 proxy (Low05)	12	737.515593	255.946024	2059.552374	2115.950916	2144.068963	0.333333	0.333333

APPENDIX F — Supplementary Exploratory Visualisations

Supplementary visualisations supporting exploratory analysis are presented below. Figures included here were not repeated in the main chapters.

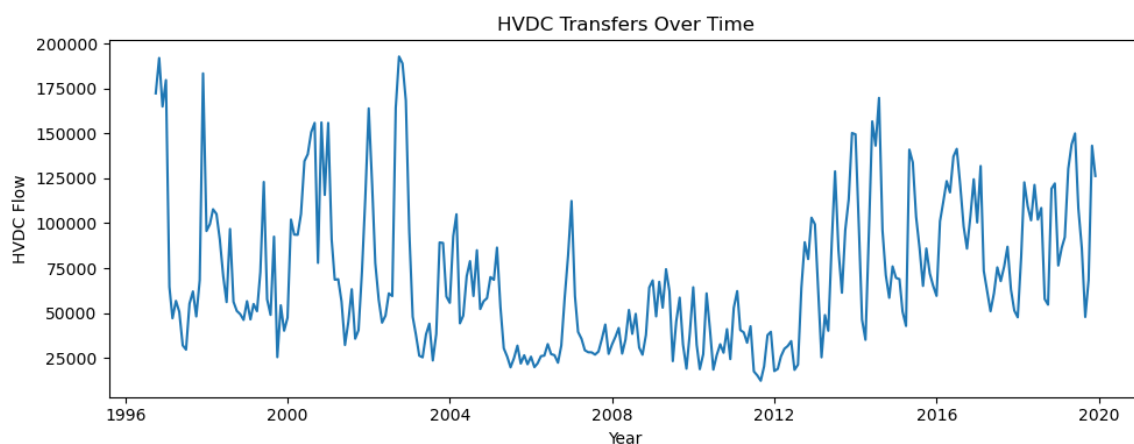


Figure F1 — Time Series of Monthly HVDC Inter-Island Transfer Flows

The figure illustrates the evolution of HVDC transfer volumes over time, highlighting periods of elevated inter-island flow that may amplify system stress during hydro scarcity conditions.

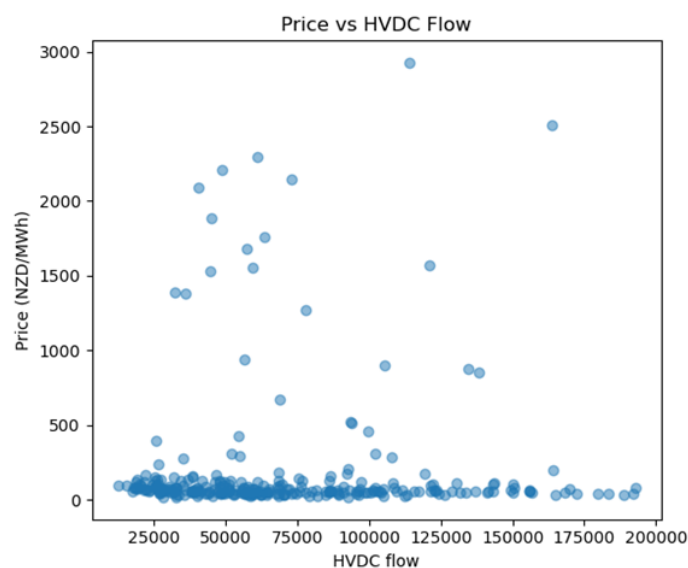


Figure F2 — Relationship Between Wholesale Electricity Prices and HVDC Inter-Island Transfer Flows

The scatterplot shows that extreme price spikes can coincide with elevated HVDC transfer levels, suggesting that inter-island flow may act as a stress amplifier during scarcity conditions rather than a primary price driver.

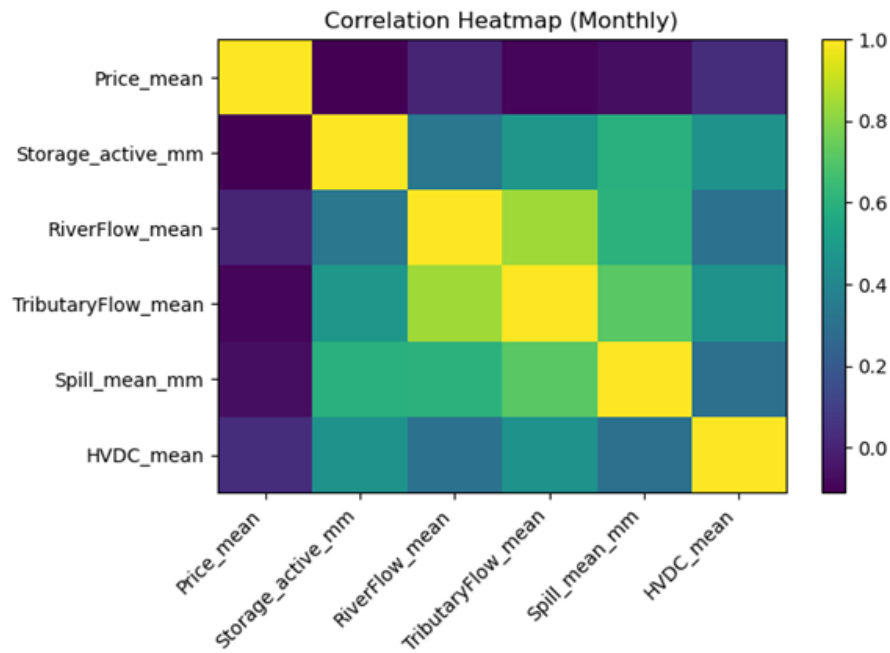


Figure F3 — Correlation Matrix of Monthly Hydro-Risk Variables

The heatmap illustrates strong positive correlations among hydrological variables (storage, inflows, spill), while wholesale prices exhibit weak linear correlation with these fundamentals, indicating potential non-linear scarcity effects.

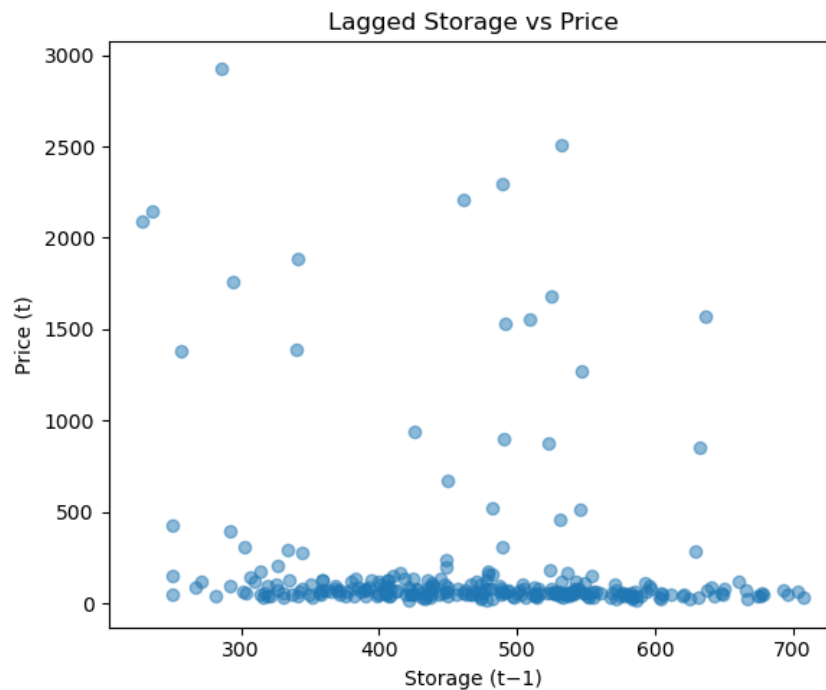


Figure F4 — Lagged Storage–Price Relationship (Storage $t-1$ vs Price t)

The scatterplot indicates that lower storage levels in the previous month are associated with a higher probability of extreme price outcomes, supporting the inclusion of lagged storage in the driver model specification.

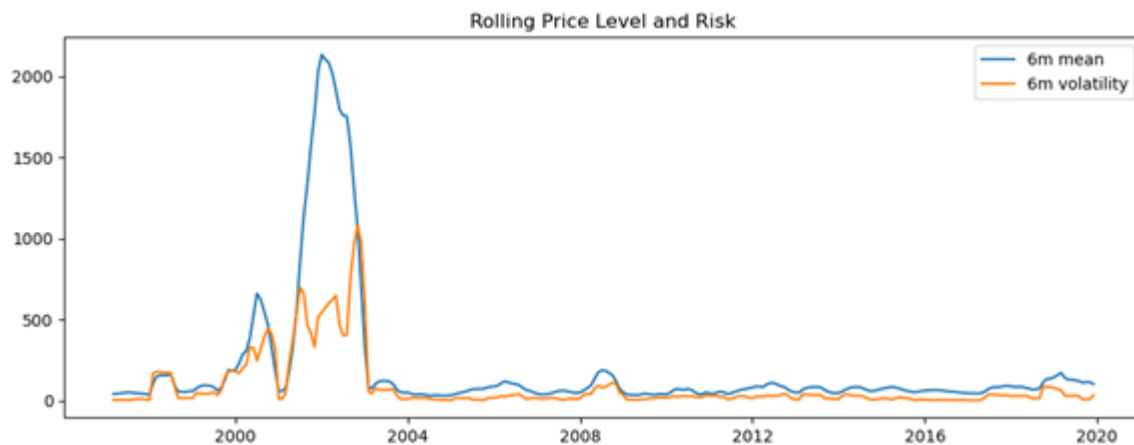


Figure F5 — Six-Month Rolling Average and Volatility of Wholesale Electricity Prices

The rolling statistics highlight periods of elevated price volatility coinciding with historical scarcity events, particularly during the early-2000s crisis, reinforcing the clustering of tail-risk episodes.

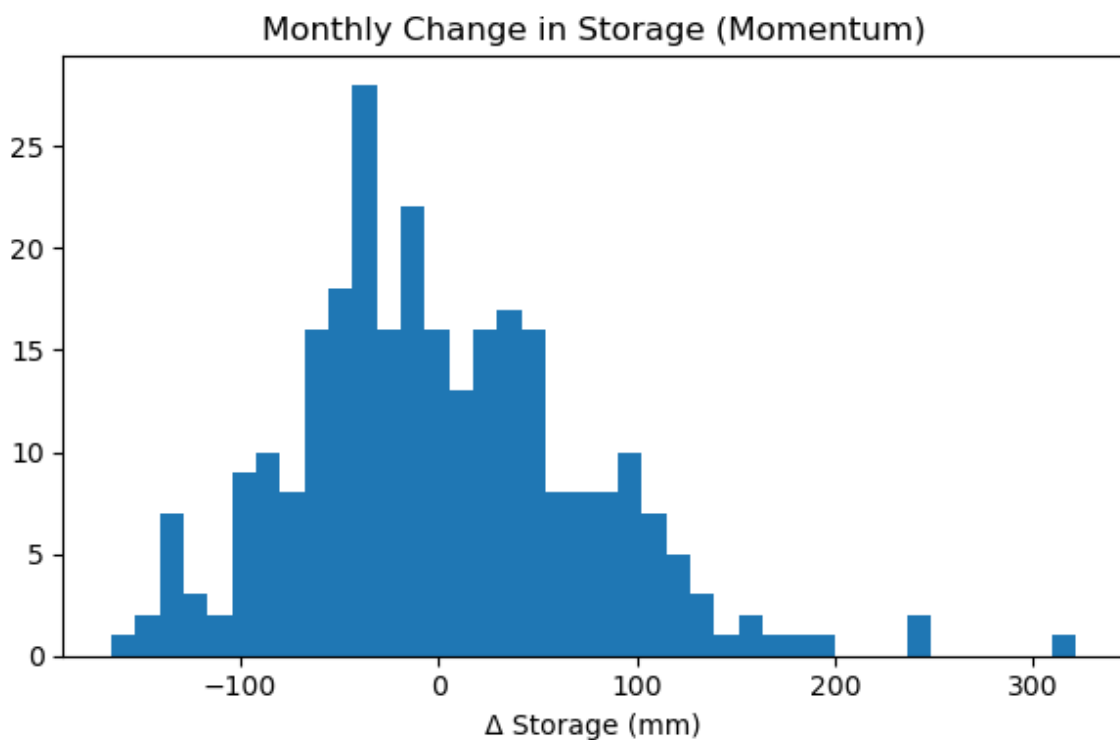


Figure F6 — Distribution of Monthly Changes in Hydro Storage (Δ Storage)

The distribution of monthly storage changes highlights periods of rapid depletion and recovery, which may act as early warning signals for emerging hydro scarcity conditions.

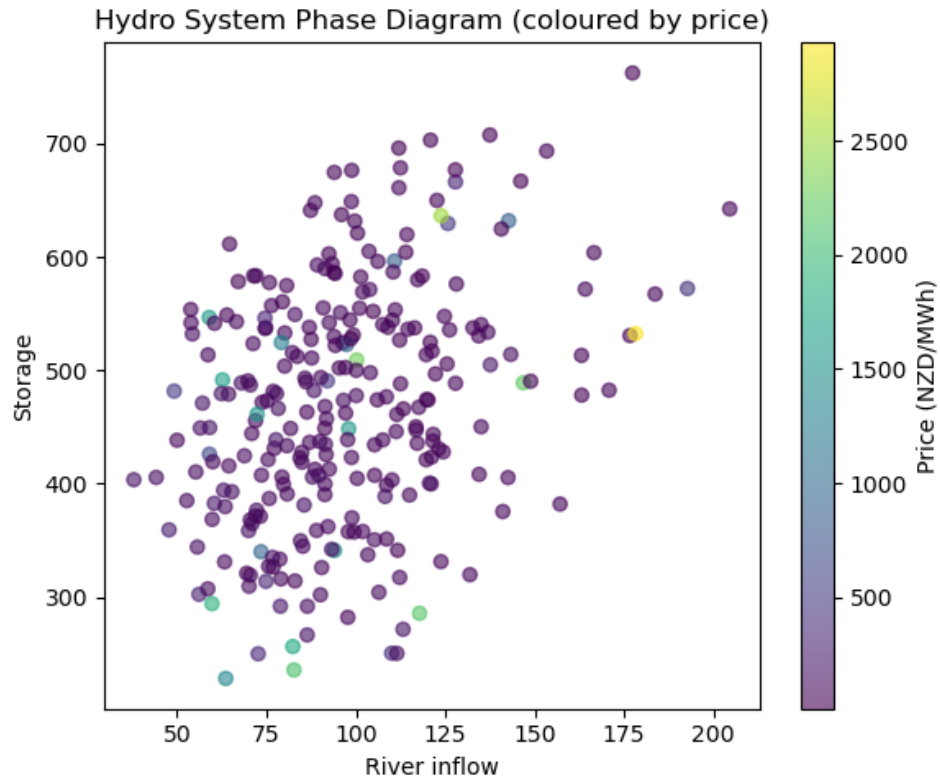


Figure F7 — Hydro System State Space: Storage and River Inflow with Price Intensity Overlay

The phase diagram illustrates system states defined by inflow and storage levels, with higher price intensities clustering in lower-storage or transition regions, supporting the scarcity-based pricing hypothesis.

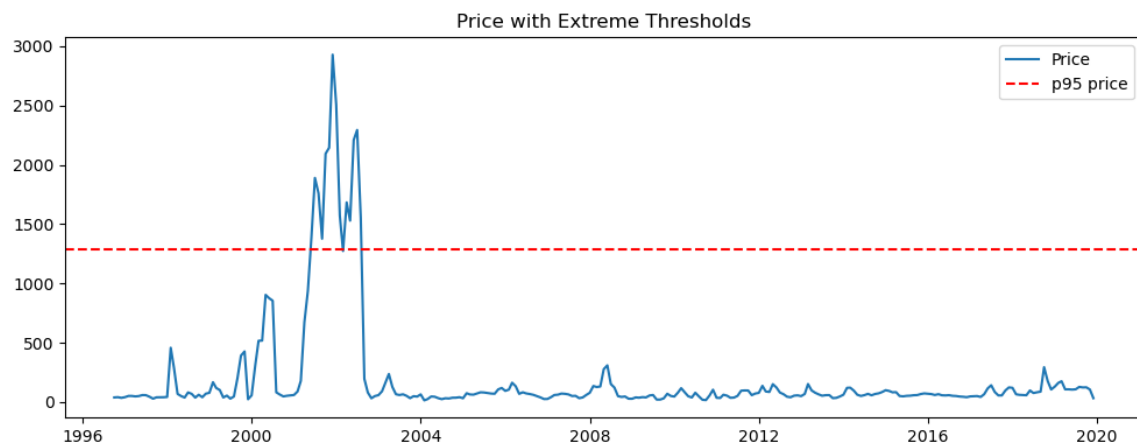


Figure F8 — Wholesale Price Series with 95th Percentile Extreme Threshold

The dashed threshold line identifies extreme price events above the 95th percentile, highlighting the clustering of tail-risk episodes during historical hydro scarcity periods.

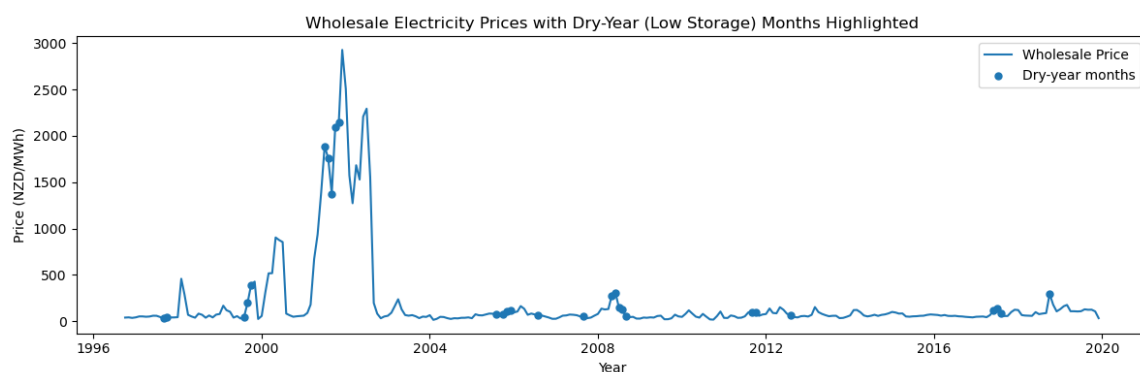


Figure F9 — Wholesale Electricity Prices with Dry-Year (Low Storage) Months Highlighted

Highlighted dry-year months coincide with elevated price levels and clustered spikes, reinforcing the regime-dependent nature of wholesale electricity risk under hydro scarcity conditions.

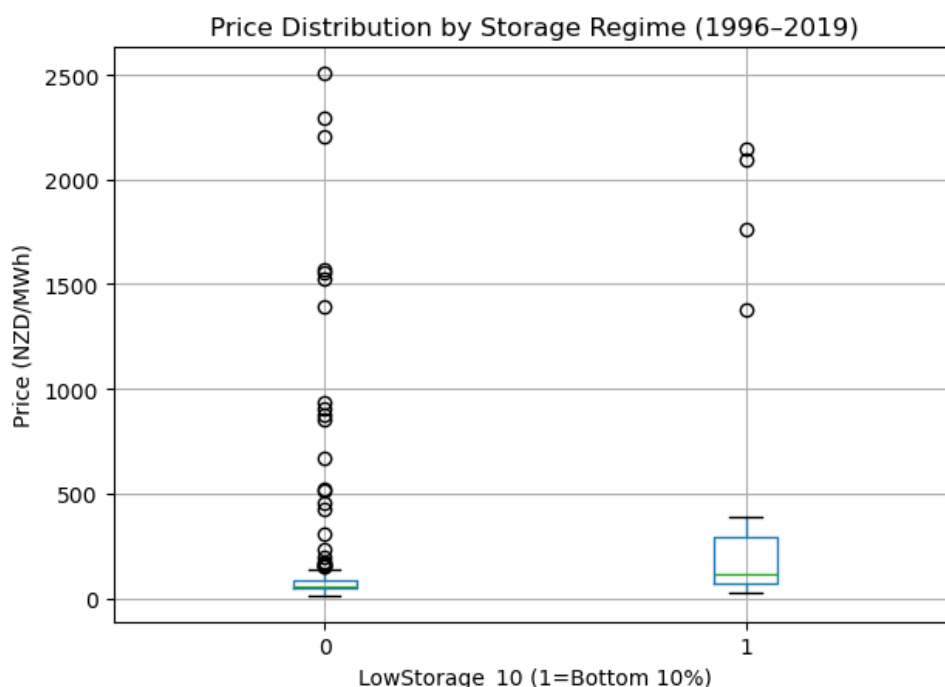


Figure F10 — Price Distribution by Storage Regime (LowStorage_10 Indicator)

The boxplot demonstrates a pronounced upward shift in both median price and tail dispersion during low-storage months, confirming the regime-dependent amplification of wholesale price risk.

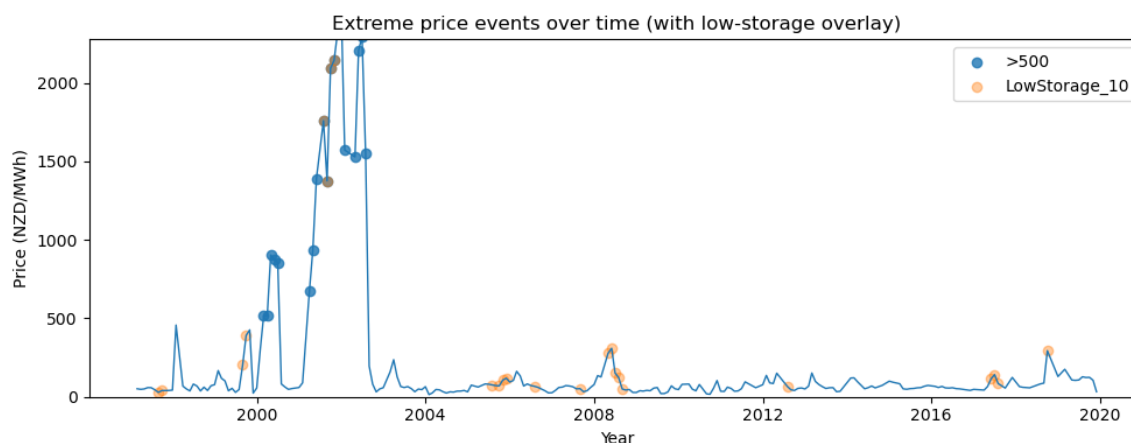


Figure F11 — Extreme Price Events Over Time with Low-Storage Overlay

Extreme price events (above the defined threshold) cluster during low-storage months, reinforcing the scarcity-driven mechanism underlying wholesale price spikes.

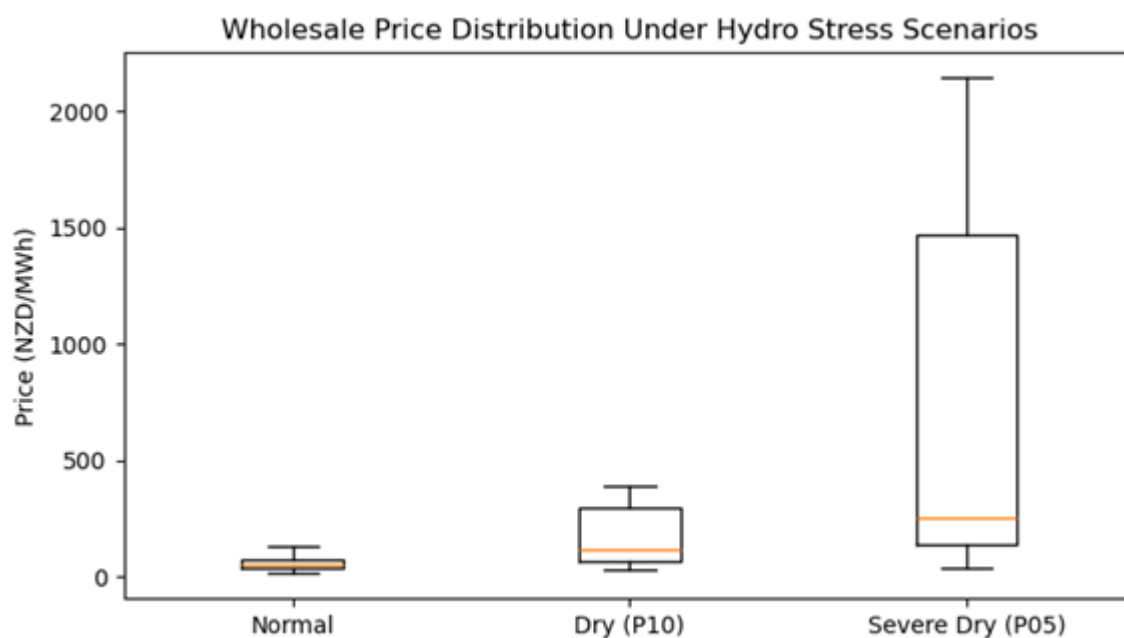


Figure F12 — Wholesale Price Distribution Under Hydro Stress Scenarios

The boxplot illustrates a substantial upward shift in both median price and tail dispersion under dry and severe dry scenarios, highlighting the asymmetric risk exposure faced during hydro scarcity events.

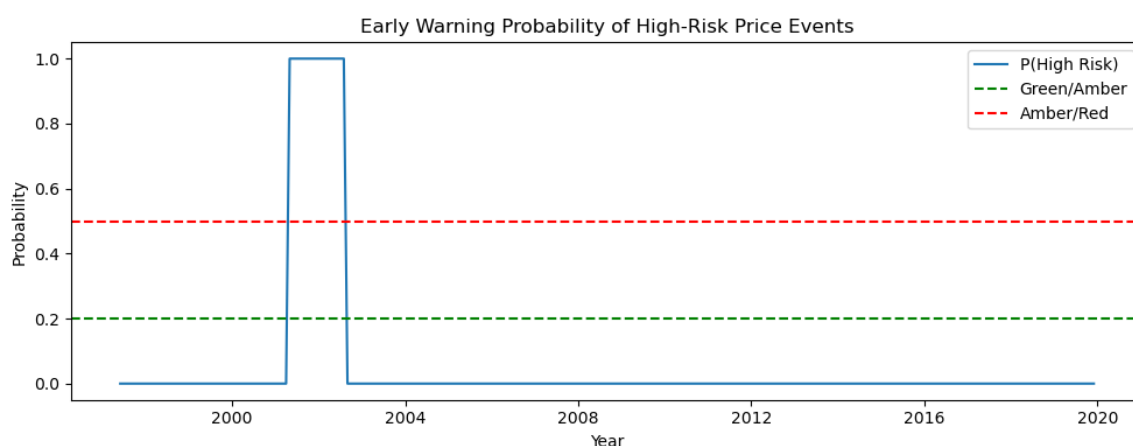


Figure F13 — Model-Implied Probability of High-Risk Wholesale Price Events with Alert Thresholds

The probability signal breaches the amber and red alert thresholds during historical scarcity periods, demonstrating the model’s potential use as an early-warning decision support tool.

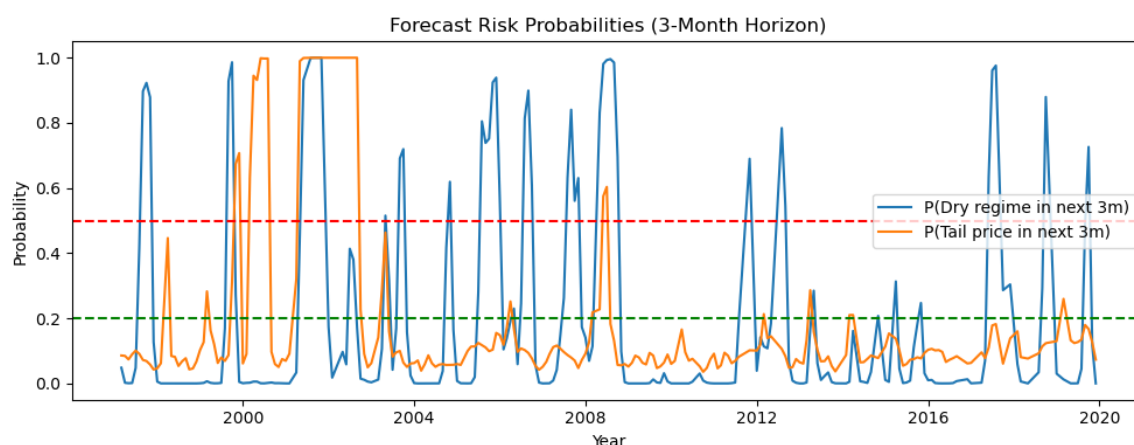


Figure F14 — Three-Month Ahead Forecast Probabilities of Dry Regime and Tail Price Events

The forecast probabilities capture forward-looking risk signals, with elevated dry-regime probabilities preceding increases in predicted tail-price risk, supporting proactive risk management planning.

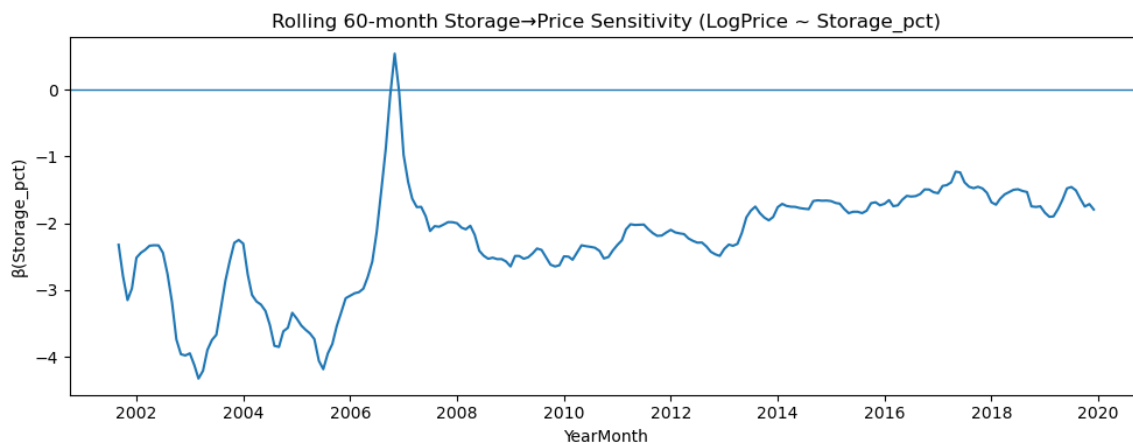


Figure F15 — Rolling 60-Month Storage–Price Sensitivity (LogPrice ~ Storage_pct)

The rolling coefficient indicates persistent negative sensitivity of prices to storage levels, with periods of heightened elasticity during historical scarcity episodes, suggesting structural shifts in hydro-risk transmission.

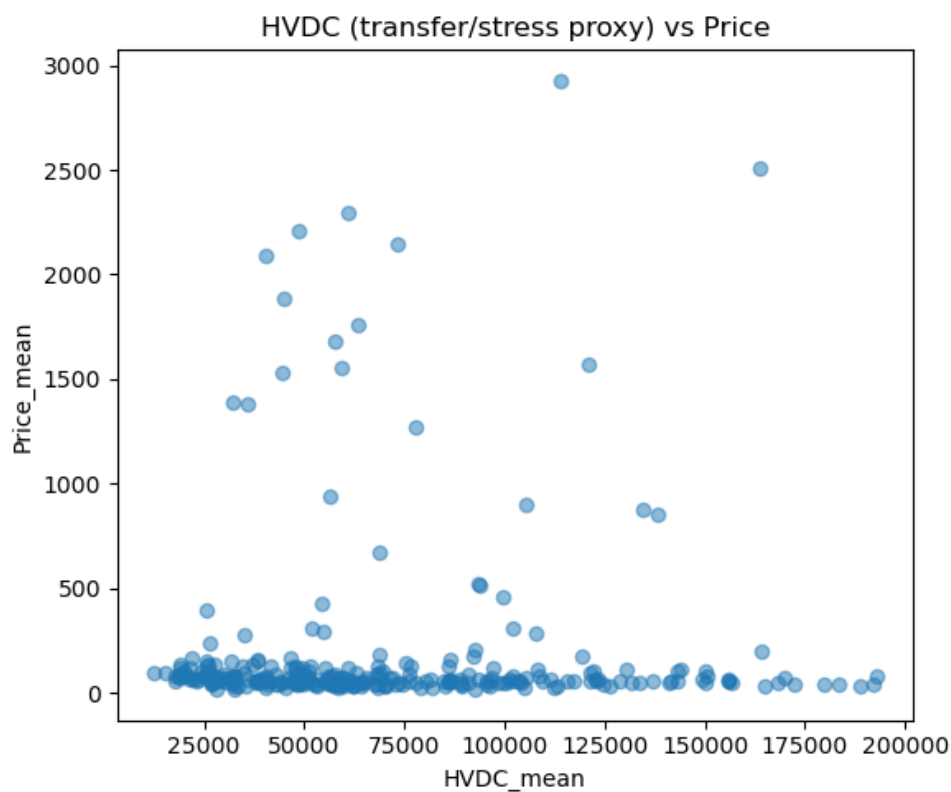


Figure F16 — Relationship Between Monthly HVDC Transfer Levels and Wholesale Electricity Prices

The scatter suggests weak linear correlation overall, but extreme price outcomes tend to occur alongside elevated HVDC transfers, indicating potential stress amplification during tight system conditions.

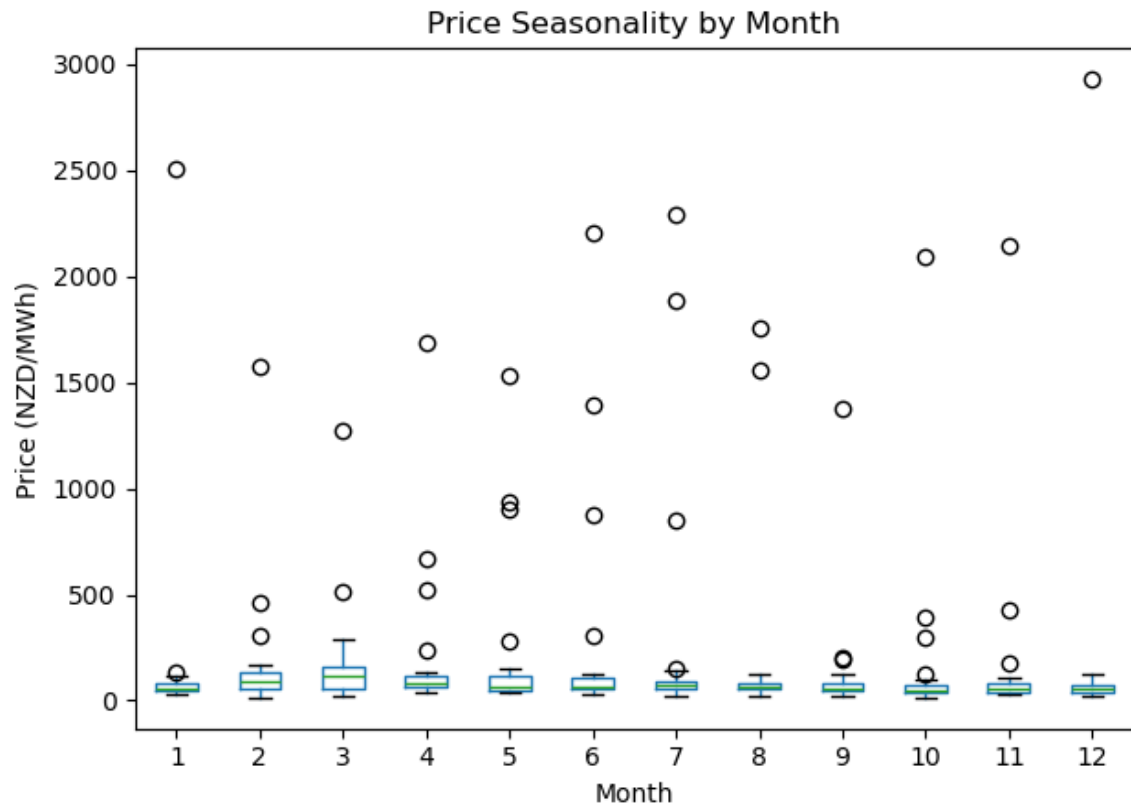


Figure F17 — Seasonal Distribution of Wholesale Electricity Prices by Calendar Month

The distribution indicates moderate seasonal variation, with greater dispersion and more extreme outliers observed in winter months. This supports the inclusion of seasonal controls in risk modelling.

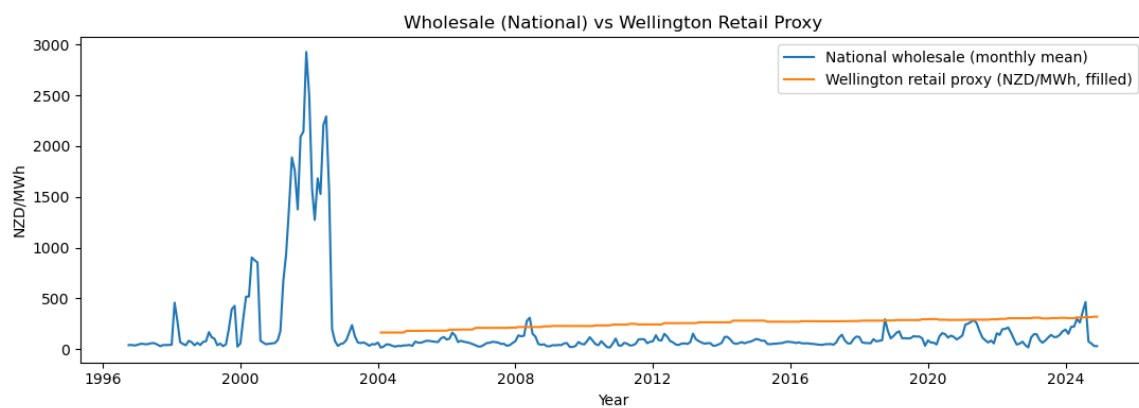


Figure F18 — National Wholesale Electricity Prices versus Wellington Retail Price Proxy

Wholesale prices exhibit substantial volatility and extreme spikes, while the retail proxy follows a smoother, upward trend. This highlights retail price stickiness and the risk transmission gap between wholesale shocks and retail pricing.

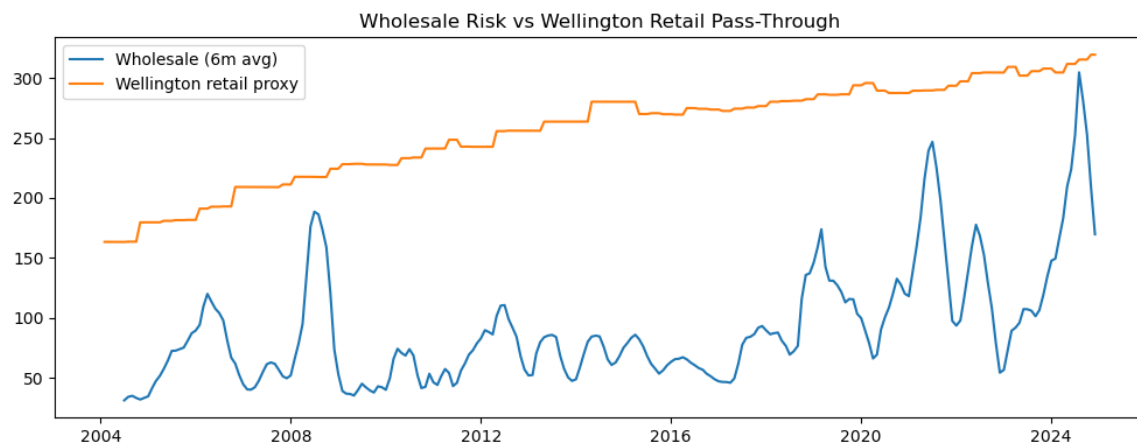


Figure F19 — Comparison of Smoothed Wholesale Risk and Wellington Retail Price Pass-Through

The 6-month wholesale average exhibits pronounced cyclical risk spikes, whereas the retail proxy adjusts gradually. This supports evidence of delayed and partial pass-through from wholesale volatility to retail pricing.

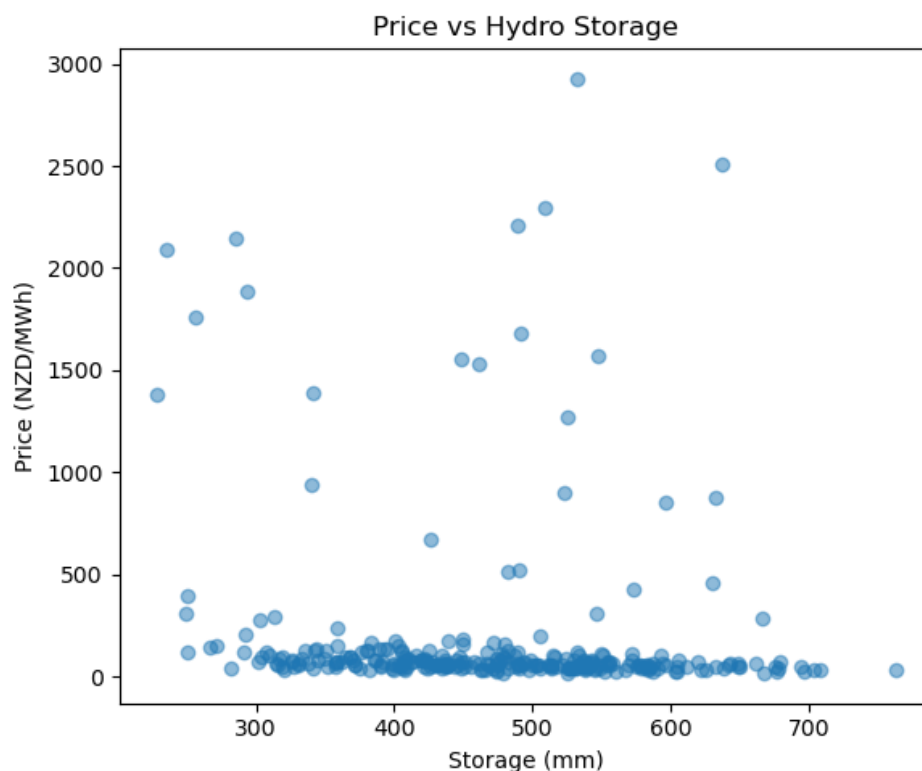


Figure F20 — Wholesale Electricity Price versus National Hydro Storage Level

The scatterplot illustrates a clear non-linear inverse relationship, with extreme price spikes concentrated at lower storage levels. This supports the scarcity-based pricing hypothesis central to the hydro-risk framework

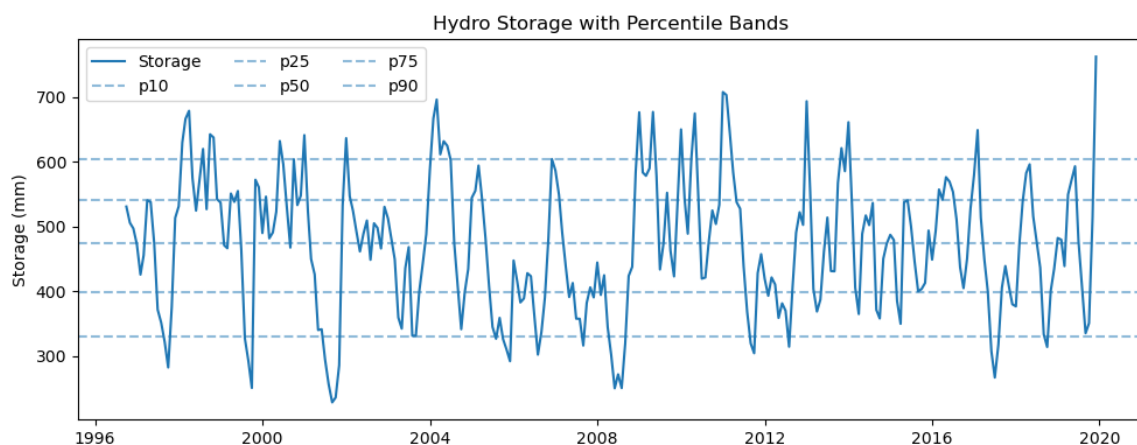


Figure F21 — Hydro Storage Levels with Historical Percentile Bands (p10–p90)

The percentile bands contextualise storage conditions relative to historical distribution, enabling objective identification of low-storage (dry) and high-storage (abundance) regimes used in regime-based risk modelling.

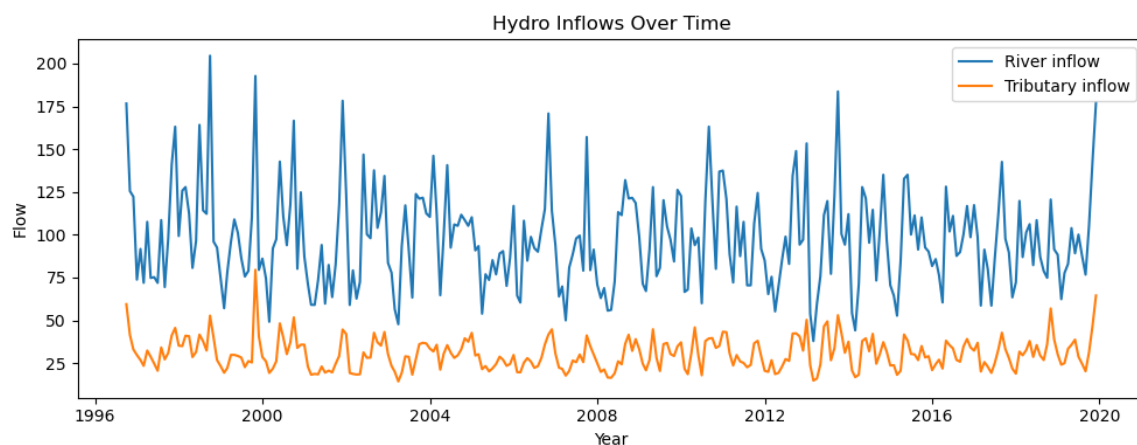


Figure F22 — River and Tributary Hydro Inflows Over Time

Inflows exhibit substantial temporal variability, with episodic surges and drought periods. These dynamics underpin storage fluctuations and contribute to the formation of hydro-scarcity regimes.

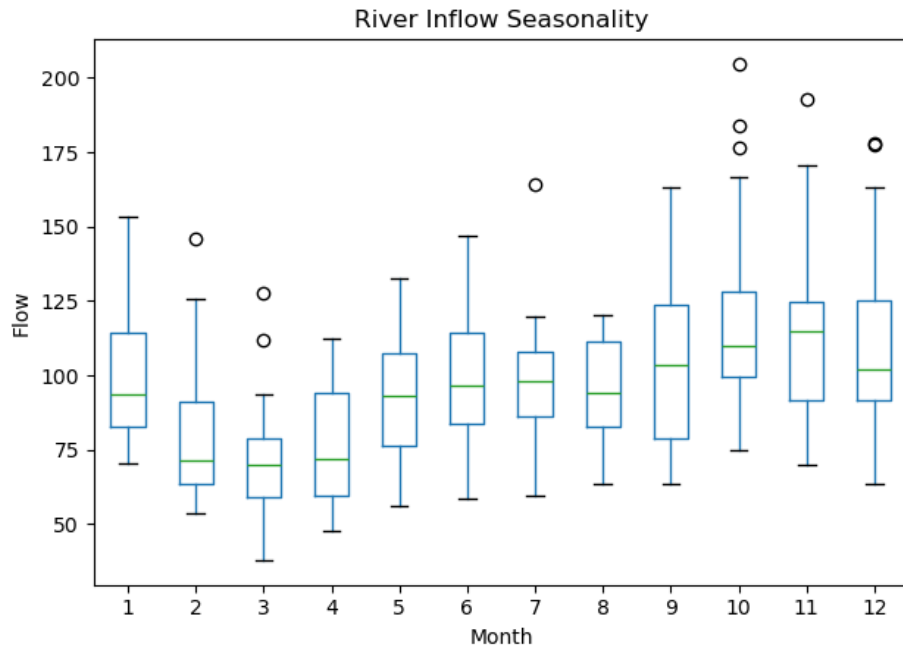


Figure F23 — Seasonal Distribution of River Inflows by Month

River inflows display clear seasonal structure, with higher median inflows during late winter and spring months. This seasonality influences storage rebuilding dynamics and the timing of hydro-related price risk.

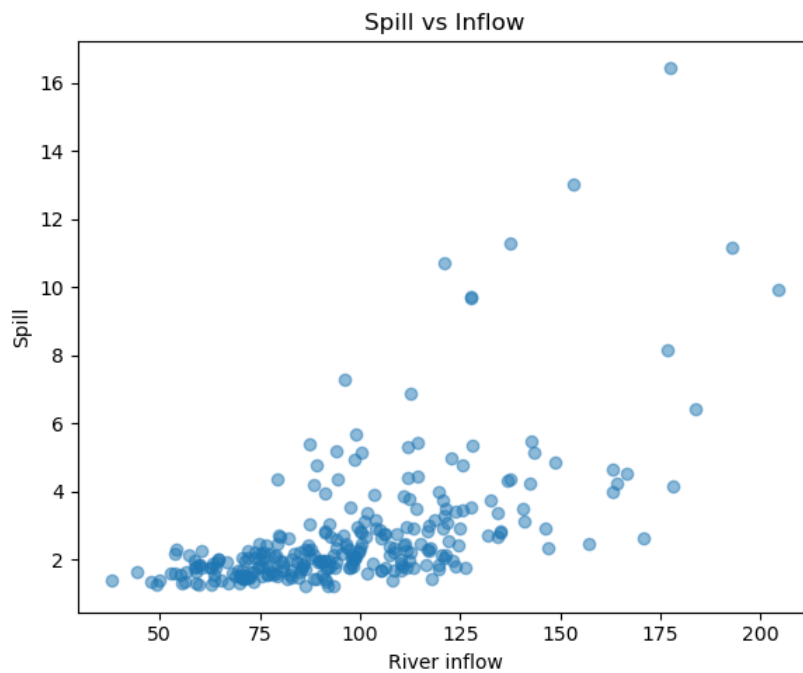


Figure F24 — Spill Dynamics as a Function of River Inflows

Spill volumes increase non-linearly with higher inflow levels, indicating capacity constraints once storage approaches upper bounds. This supports the interpretation of spill as an abundance-state indicator in regime classification.

APPENDIX G — Diagnostics

G.1 Residual and Fit Diagnostics

Regression diagnostics are provided below to support model validity. The Omnibus and Jarque-Bera (JB) statistics indicate non-normal residuals, consistent with heavy-tailed price behaviour.

Table G1 — OLS Model Summary (Fit Statistics)

Dep. Variable:	LogPrice	R-squared:	0.167
Model:	OLS	Adj. R-squared:	0.157
Method:	Least Squares	F-statistic:	11.95
Date:	Sun, 15 Feb 2026	Prob (F-statistic):	2.63e-07
Time:	19:52:44	Log-Likelihood:	-307.72
No. Observations:	238	AIC:	623.4
Df Residuals:	234	BIC:	637.3
Df Model:	3		
Covariance Type:	HC3		

Table G2 — OLS Coefficients (Robust SE: HC3)

	coef	std err	z	P> z	[0.025	0.975]
Intercept	5.0764	0.318	15.954	0.000	4.453	5.700
Storage_pct	-1.1764	0.500	-2.351	0.019	-2.157	-0.196
LowStorage_10	8.3456	1.719	4.854	0.000	4.976	11.715
Storage_pct:LowStorage_10	-20.6417	4.214	-4.898	0.000	-28.901	-12.382

Table G3 — OLS Diagnostics

Omnibus:	109.162	Durbin-Watson:	0.477
Prob(Omnibus):	0.0	Jarque-Bera (JB):	374.175
Skew:	1.993	Prob(JB):	5.61e-82
Kurtosis:	7.674	Cond. No.	98.9

Table G4 — OLS Model Summary (Fit Statistics)

Dep. Variable:	LogPrice	R-squared:	0.705
Model:	OLS	Adj. R-squared:	0.698
Method:	Least Squares	F-statistic:	75.06
Date:	Sun, 15 Feb 2026	Prob (F-statistic):	1.56e-46
Time:	19:52:45	Log-Likelihood:	-184.43
No. Observations:	238	AIC:	380.9
Df Residuals:	232	BIC:	401.7
Df Model:	5		
Covariance Type:	HC3		

Table G5 — OLS Coefficients (Robust SE: HC3)

	coef	std err	z	P> z	[0.025	0.975]
Intercept	4.4830	0.195	22.992	0.000	4.101	4.865
Storage_pct	-0.8740	0.360	-2.425	0.015	-1.580	-0.168
HVDC_mean	1.516e-06	1.02e-06	1.490	0.136	-4.78e-07	3.51e-06
Spill_mean_mm	0.0155	0.027	0.583	0.560	-0.037	0.068
Inflow_z_lag1	-0.2749	0.044	-6.302	0.000	-0.360	-0.189
Price_lag1	0.0017	0.000	15.532	0.000	0.001	0.002

Table G6 — OLS Diagnostics

Omnibus:	37.695	Durbin-Watson:	1.173
Prob(Omnibus):	0.0	Jarque-Bera (JB):	69.579
Skew:	0.828	Prob(JB):	7.78e-16
Kurtosis:	5.067	Cond. No.	873000.0

Table G7 — OLS Model Summary (Fit Statistics)

Dep. Variable:	LogPrice	R-squared:	0.708
Model:	OLS	Adj. R-squared:	0.701
Method:	Least Squares	F-statistic:	62.54
Date:	Sun, 15 Feb 2026	Prob (F-statistic):	1.0599999999999999e-45
Time:	19:52:47	Log-Likelihood:	-182.94
No. Observations:	238	AIC:	379.9
Df Residuals:	231	BIC:	404.2
Df Model:	6		
Covariance Type:	HC3		

Table G8 — OLS Coefficients (Robust SE: HC3)

	coef	std err	z	P> z	[0.025	0.975]
Intercept	4.3002	0.215	19.979	0.000	3.878	4.722
Storage_pct	-0.5431	0.420	-1.293	0.196	-1.366	0.280
Storage_below_knot	3.7226	2.543	1.464	0.143	-1.261	8.706
HVDC_mean	1.409e-06	1.01e-06	1.392	0.164	-5.76e-07	3.39e-06
Spill_mean_mm	0.0063	0.029	0.220	0.826	-0.050	0.062
Inflow_z_lag1	-0.2740	0.044	-6.291	0.000	-0.359	-0.189
Price_lag1	0.0017	0.000	14.093	0.000	0.001	0.002

Table G9 — OLS Diagnostics

Omnibus:	39.479	Durbin-Watson:	1.171
Prob(Omnibus):	0.0	Jarque-Bera (JB):	74.078
Skew:	0.858	Prob(JB):	8.21e-17
Kurtosis:	5.128	Cond. No.	5000000.0

Table G10 — OLS Model Summary (Fit Statistics)

Dep. Variable:	LogPrice	R-squared:	0.711
Model:	OLS	Adj. R-squared:	0.699
Method:	Least Squares	F-statistic:	38.85
Date:	Sun, 15 Feb 2026	Prob (F-statistic):	2.4e-41
Time:	19:52:47	Log-Likelihood:	-181.85
No. Observations:	238	AIC:	383.7
Df Residuals:	228	BIC:	418.4
Df Model:	9		
Covariance Type:	HC3		

Table G11 — OLS Coefficients (Robust SE: HC3)

	coef	std err	z	P> z	[0.025	0.975]
Intercept	4.6083	0.425	10.843	0.000	3.775	5.441
bs(Storage_pct, df=5, degree=3, include intercept=False)[0]	-0.6073	0.621	-0.978	0.328	-1.824	0.609
bs(Storage_pct, df=5, degree=3, include intercept=False)[1]	-0.6328	0.361	-1.751	0.080	-1.341	0.075
bs(Storage_pct, df=5, degree=3, include intercept=False)[2]	-0.8390	0.534	-1.571	0.116	-1.886	0.208
bs(Storage_pct, df=5, degree=3, include intercept=False)[3]	-0.6424	0.644	-0.997	0.319	-1.905	0.620
bs(Storage_pct, df=5, degree=3, include intercept=False)[4]	-1.8636	1.455	-1.281	0.200	-4.715	0.988
HVDC_mean	1.363e-06	1.15e-06	1.180	0.238	-9.01e-07	3.63e-06
Spill mean mm	0.0343	0.055	0.622	0.534	-0.074	0.142
Inflow_z_lag1	-0.2818	0.046	-6.174	0.000	-0.371	-0.192
Price lag1	0.0017	0.000	13.838	0.000	0.001	0.002

Table G12 — OLS Diagnostics

Omnibus:	35.426	Durbin-Watson:	1.159
Prob(Omnibus):	0.0	Jarque-Bera (JB):	63.391

Skew:	0.795	Prob(JB):	1.72e-14
Kurtosis:	4.966	Cond. No.	2030000.0

Table G13 — OLS Model Summary (Fit Statistics)

Dep. Variable:	LogPrice	R-squared:	0.08
Model:	OLS	Adj. R-squared:	0.07
Method:	Least Squares	F-statistic:	25.22
Date:	Sun, 15 Feb 2026	Prob (F-statistic):	1.91e-14
Time:	19:53:15	Log-Likelihood:	-379.64
No. Observations:	279	AIC:	767.3
Df Residuals:	275	BIC:	781.8
Df Model:	3		
Covariance Type:	HC3		

Table G14 — OLS Coefficients (Robust SE: HC3)

	coef	std err	z	P> z	[0.025	0.975]
Intercept	5.6658	0.438	12.934	0.000	4.807	6.524
Storage_pct	-1.8359	0.690	-2.660	0.008	-3.189	-0.483
PostBreak	-0.2551	0.464	-0.549	0.583	-1.165	0.655
Storage_pct:PostBreak	0.0230	0.729	0.032	0.975	-1.405	1.451

Table G15 — OLS Diagnostics

Omnibus:	104.335	Durbin-Watson:	0.274
Prob(Omnibus):	0.0	Jarque-Bera (JB):	277.879
Skew:	1.754	Prob(JB):	4.56e-61
Kurtosis:	6.405	Cond. No.	24.8

G.2 Justification for HC3 Robust Standard Errors

HC3 robust standard errors were used to reduce sensitivity to heteroskedasticity and heavy-tailed residuals, which are expected under electricity price spikes and dry-year scarcity conditions. Robust inference helps prevent overstating coefficient significance when variance is non-constant.

APPENDIX H — Reproducibility and Code Extracts

H.1 Data Preparation Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import statsmodels.api as sm
import statsmodels.formula.api as smf

from pathlib import Path
import warnings
warnings.filterwarnings("ignore")

pd.set_option("display.max_columns", 250)
pd.set_option("display.width", 220)

print("Environment initialised successfully.")

BASE_DIR = Path(r"C:\Users\sidhu\Desktop\Final project files sorted")
DATA_DIR = BASE_DIR

INFRA_DIR = DATA_DIR / "infrastructure and hydroconstraints Attributes"
DERIVED_ENERGY_DIR = DATA_DIR / "DerivedSeriesForStorageCapacityAndSpecificEnergy"
FLOWS_DIR = DATA_DIR / "Flows"
TRIBUTARY_DIR = DATA_DIR / "DerivedTributaryFlows"
STORAGE_DIR = DATA_DIR / "Storage"
SPILL_DIR = DATA_DIR / "spill"
PRICES_DIR = DATA_DIR / "Final prices Monthly"
HVDC_DIR = DATA_DIR / "HVDCflows data"
CLIMATE_DIR = DATA_DIR / "NIWA_Climate_Monthly"

OUTPUT_DIR = BASE_DIR / "outputs_notebook"
OUTPUT_DIR.mkdir(exist_ok=True)

FIG_DIR = OUTPUT_DIR / "figures"
FIG_DIR.mkdir(exist_ok=True)

paths = {
    "INFRA_DIR": INFRA_DIR,
    "DERIVED_ENERGY_DIR": DERIVED_ENERGY_DIR,
    "FLOWS_DIR": FLOWS_DIR,
    "TRIBUTARY_DIR": TRIBUTARY_DIR,
    "STORAGE_DIR": STORAGE_DIR,
    "SPILL_DIR": SPILL_DIR,
    "PRICES_DIR": PRICES_DIR,
    "HVDC_DIR": HVDC_DIR,
    "CLIMATE_DIR": CLIMATE_DIR
}

path_check = pd.DataFrame({
    "path_key": list(paths.keys()),
    "path": [str(p) for p in paths.values()],
    "exists": [p.exists() for p in paths.values()],
    "is_dir": [p.is_dir() for p in paths.values()],
})
```


path_check

```
def audit_folder(path: Path) -> pd.DataFrame:
    files = [f for f in path.iterdir() if f.is_file()]
    return pd.DataFrame({
        "folder": path.name,
        "n_files": len(files),
        "extensions": [sorted({f.suffix.lower() for f in files})]
    })

inventory_summary = pd.concat([audit_folder(p) for p in paths.values()], ignore_index=True)
inventory_summary

def standardise_columns(df: pd.DataFrame) -> pd.DataFrame:
    df = df.copy()
    df.columns = (
        df.columns.astype(str)
        .str.strip()
        .str.lower()
        .str.replace(r"\s+", "_", regex=True)
        .str.replace(r"^[a-z0-9_]", "", regex=True)
    )
    return df

def read_data_file(path: Path) -> pd.DataFrame | None:
    try:
        if path.suffix.lower() == ".csv":
            df = pd.read_csv(path)
            return standardise_columns(df)

        if path.suffix.lower() in [".xlsx", ".xls"]:
            xls = pd.ExcelFile(path)
            for sheet in xls.sheet_names:
                df = pd.read_excel(xls, sheet_name=sheet)
                if df.shape[0] > 10 and df.shape[1] > 2:
                    return standardise_columns(df)
            return None
    except Exception:
        return None

def load_folder(folder: Path, label: str) -> pd.DataFrame:
    files = [f for f in folder.iterdir() if f.is_file()]
    tables = []
    for f in files:
        df = read_data_file(f)
        if df is None or df.empty:
            continue
        df["source_file"] = f.name
        df["source_folder"] = label
        tables.append(df)

    if not tables:
        return pd.DataFrame()
    return pd.concat(tables, ignore_index=True, sort=False)

infrastructure_attributes = load_folder(INFRA_DIR, "infrastructure")
derived_energy = load_folder(DERIVED_ENERGY_DIR, "derived_energy")
```

```

river_flows = load_folder(FLOWS_DIR, "flows")
tributary_flows = load_folder(TRIBUTARY_DIR, "tributary_flows")
storage_levels = load_folder(STORAGE_DIR, "storage")
spill_data = load_folder(SPILL_DIR, "spill")
prices_raw = load_folder(PRICES_DIR, "prices")
hvdc_raw = load_folder(HVDC_DIR, "hvdc")
climate_raw = load_folder(CLIMATE_DIR, "climate")

{k: v.shape for k, v in {
    "infrastructure": infrastructure_attributes,
    "derived_energy": derived_energy,
    "river_flows": river_flows,
    "tributary_flows": tributary_flows,
    "storage_levels": storage_levels,
    "spill_data": spill_data,
    "prices_raw": prices_raw,
    "hvdc_raw": hvdc_raw,
    "climate_raw": climate_raw
}.items()}}

def peek(df: pd.DataFrame, name: str, n=5):
    print(f"\n{name} - shape: {df.shape}")
    display(df.head(n))
    print("Columns:", list(df.columns)[:25], "..." if df.shape[1] > 25 else "")

peek(prices_raw, "prices_raw")
peek(hvdc_raw, "hvdc_raw")
peek(storage_levels, "storage_levels")
peek(river_flows, "river_flows")
peek(spill_data, "spill_data")

# =====
# 2.3 Date Parsing & Time Alignment (MEMORY SAFE)
# =====

def ensure_datetime_inplace(df: pd.DataFrame, col: str, dayfirst: bool = False) ->
pd.DataFrame:
    """
    Convert ONE column to datetime WITHOUT copying the full dataframe.
    This is critical for huge tables (e.g., prices_raw).
    """
    if col in df.columns and not np.issubdtype(df[col].dtype, np.datetime64):
        df[col] = pd.to_datetime(df[col], errors="coerce", dayfirst=dayfirst)
    return df

def add_yearmonth_inplace(df: pd.DataFrame, date_col: str, out_col: str = "YearMonth") ->
pd.DataFrame:
    """
    Add YearMonth WITHOUT copying full df.
    """
    if date_col in df.columns:
        df[out_col] = df[date_col].dt.to_period("M").dt.to_timestamp()
    return df

```

```

# Prices (HUGE) - do NOT copy

# Prefer 'tradingdate' if present; else fallback to 'trading_date'
if "tradingdate" in prices_raw.columns:
    prices_raw = ensure_datetime_inplace(prices_raw, "tradingdate", dayfirst=False)
    prices_raw = add_yearmonth_inplace(prices_raw, "tradingdate")
elif "trading_date" in prices_raw.columns:
    prices_raw = ensure_datetime_inplace(prices_raw, "trading_date", dayfirst=False)
    prices_raw = add_yearmonth_inplace(prices_raw, "trading_date")

# HVDC - dayfirst strings like 01/10/1996

if "trading_date" in hvdc_raw.columns:
    hvdc_raw = ensure_datetime_inplace(hvdc_raw, "trading_date", dayfirst=True)
    hvdc_raw = add_yearmonth_inplace(hvdc_raw, "trading_date")
elif "tradingdate" in hvdc_raw.columns:
    hvdc_raw = ensure_datetime_inplace(hvdc_raw, "tradingdate", dayfirst=True)
    hvdc_raw = add_yearmonth_inplace(hvdc_raw, "tradingdate")

# Hydro datasets (smaller) - safe inplace too

for d, datecol in [
    (river_flows, "date"),
    (tributary_flows, "date"),
    (storage_levels, "date"),
    (spill_data, "date"),
]:
    ensure_datetime_inplace(d, datecol, dayfirst=False)
    add_yearmonth_inplace(d, datecol)

# Climate (monthly anchor datemin)
if "datemin" in climate_raw.columns:
    climate_raw = ensure_datetime_inplace(climate_raw, "datemin", dayfirst=False)
    climate_raw = add_yearmonth_inplace(climate_raw, "datemin")

START_DATE = pd.Timestamp("1996-10-01")
END_FULL_SYSTEM = pd.Timestamp("2019-12-01") # Last year with HVDC coverage
END_EXTENDED = pd.Timestamp("2024-12-01") # Last year with hydro/flows/storage/spill coverage

def filter_monthly(df_m: pd.DataFrame, start: pd.Timestamp, end: pd.Timestamp) ->
pd.DataFrame:
    # monthly df only (small) -> safe to copy
    return df_m.loc[(df_m["YearMonth"] >= start) & (df_m["YearMonth"] <= end)].copy()

# PRICES: ultra memory-safe monthly aggregation (FIXED)

# unify date col name
if "tradingdate" not in prices_raw.columns and "trading_date" in prices_raw.columns:
    prices_raw = prices_raw.rename(columns={"trading_date": "tradingdate"})

```

```

td = pd.to_datetime(prices_raw["tradingdate"], errors="coerce", cache=True)
px = prices_raw["dollarspermegawatthour"]

# month key (numpy month)
ym = td.values.astype("datetime64[M]")

# groupby month key (NaT auto dropped)
prices_monthly_all = (
    pd.Series(px.to_numpy(), name="Price_mean")
        .groupby(ym)
        .mean()
        .reset_index()
        .rename(columns={"index": "YearMonth"})
)

# convert the small monthly output to month-start Timestamp
prices_monthly_all["YearMonth"] =
pd.to_datetime(prices_monthly_all["YearMonth"]).dt.to_period("M").dt.to_timestamp()
prices_monthly_all = prices_monthly_all.sort_values("YearMonth").reset_index(drop=True)

prices_monthly_all.head(), prices_monthly_all.shape

# HVDC: memory-safe monthly aggregation (1996-2019 coverage)

hv = hvdc_raw

# 1) parse trading_date (dayfirst in your file)
td = pd.to_datetime(hv["trading_date"], dayfirst=True, errors="coerce", cache=True)

# 2) month key (numpy month)
ym = td.values.astype("datetime64[M]")

# 3) TP columns
tp_cols = [c for c in hv.columns if c.startswith("tp")]
if len(tp_cols) == 0:
    raise ValueError("No TP columns found (expected tp1..tp50).")

# 4) daily mean across TP columns (compute in-place without copying whole df)
tp_vals = hv[tp_cols].to_numpy(dtype="float32", copy=False)
hvdc_daily_mean = np.nanmean(tp_vals, axis=1)

# 5) monthly mean of daily means
hvdc_monthly_all = (
    pd.Series(hvdc_daily_mean, name="HVDC_mean")
        .groupby(ym)
        .mean()
        .reset_index()
        .rename(columns={"index": "YearMonth"})
)

# 6) month-start timestamps
hvdc_monthly_all["YearMonth"] =
pd.to_datetime(hvdc_monthly_all["YearMonth"]).dt.to_period("M").dt.to_timestamp()
hvdc_monthly_all = hvdc_monthly_all.sort_values("YearMonth").reset_index(drop=True)

hvdc_monthly_all.head(), hvdc_monthly_all.shape

```

```

# Hydro monthly (flows / storage / spill)
def _monthly_mean_from_date_value(df, date_col, value_col, dayfirst=False, out_name=None):
    td = pd.to_datetime(df[date_col], errors="coerce", dayfirst=dayfirst, cache=True)
    ym = td.values.astype("datetime64[M]")
    s = pd.Series(df[value_col].to_numpy(), name=out_name or value_col)
    out = (
        s.groupby(ym).mean()
        .reset_index()
        .rename(columns={"index": "YearMonth", (out_name or value_col): (out_name or
value_col)}))
    )
    out["YearMonth"] = pd.to_datetime(out["YearMonth"]).dt.to_period("M").dt.to_timestamp()
    return out.sort_values("YearMonth").reset_index(drop=True)

flows_monthly_all = _monthly_mean_from_date_value(river_flows, "date", "flow_ms",
out_name="RiverFlow_mean")
trib_monthly_all = _monthly_mean_from_date_value(tributary_flows, "date", "flow_ms",
out_name="TributaryFlow_mean")
storage_monthly_all = _monthly_mean_from_date_value(storage_levels, "date",
"active_storage_mm", out_name="Storage_active_mm")
spill_monthly_all = _monthly_mean_from_date_value(spill_data, "date", "spill_mm",
out_name="Spill_mean_mm")

flows_monthly_all.head(), storage_monthly_all.shape

# Climate monthly
cl = climate_raw.copy()

# month key from datemin (already monthly-ish)
td = pd.to_datetime(cl["datemin"], errors="coerce", cache=True)
ym = td.values.astype("datetime64[M]")

num_cols = cl.select_dtypes(include="number").columns.tolist()

# if your numeric columns are only agentno/version (metadata), keep them but flag it
climate_monthly_all = (
    cl[num_cols]
    .assign(_ym=ym)
    .groupby("_ym", as_index=False)
    .mean()
    .rename(columns={"_ym": "YearMonth"})
)

climate_monthly_all["YearMonth"] =
pd.to_datetime(climate_monthly_all["YearMonth"]).dt.to_period("M").dt.to_timestamp()
climate_monthly_all = climate_monthly_all.sort_values("YearMonth").reset_index(drop=True)

climate_monthly_all.head(), climate_monthly_all.shape

# Build TWO PANELS
START_DATE = pd.Timestamp("1996-10-01")
END_FULL_SYSTEM = pd.Timestamp("2019-12-01")
END_EXTENDED = pd.Timestamp("2024-12-01")

def _clip_window(df, start, end):
    return df[(df["YearMonth"] >= start) & (df["YearMonth"] <=

```

```

end)].sort_values("YearMonth").reset_index(drop=True)

# Panel A: FULL system (has HVDC)
tables_full = [
    _clip_window(prices_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(hvdc_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(flows_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(trib_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(storage_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(spill_monthly_all, START_DATE, END_FULL_SYSTEM),
    _clip_window(climate_monthly_all, START_DATE, END_FULL_SYSTEM),
]

from functools import reduce
panel_full = reduce(lambda L,R: pd.merge(L, R, on="YearMonth", how="left"), tables_full)

# Panel B: EXTENDED hydro (NO HVDC)
tables_ext = [
    _clip_window(prices_monthly_all, START_DATE, END_EXTENDED),
    _clip_window(flows_monthly_all, START_DATE, END_EXTENDED),
    _clip_window(trib_monthly_all, START_DATE, END_EXTENDED),
    _clip_window(storage_monthly_all, START_DATE, END_EXTENDED),
    _clip_window(spill_monthly_all, START_DATE, END_EXTENDED),
    _clip_window(climate_monthly_all, START_DATE, END_EXTENDED),
]
panel_ext = reduce(lambda L,R: pd.merge(L, R, on="YearMonth", how="left"), tables_ext)

panel_full.shape, panel_ext.shape, panel_full.head()

```

H.2 Feature Engineering Code

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = panel_full.copy()

DATE_COL = "YearMonth"
PRICE_COL = "Price_mean"
STORAGE_COL = "Storage_active_mm"

df[DATE_COL] = pd.to_datetime(df[DATE_COL])
df = df.sort_values(DATE_COL).reset_index(drop=True)

import pandas as pd
import numpy as np

df = panel_full.copy()

STORAGE_COL = "Storage_active_mm"
PRICE_COL = "Price_mean"

# Create storage regimes using quantiles
q_low = df[STORAGE_COL].quantile(0.20)
q_high = df[STORAGE_COL].quantile(0.80)

```

```

def storage_regime(x):
    if x <= q_low:
        return "Low storage"
    elif x >= q_high:
        return "High storage"
    else:
        return "Normal storage"

df["Storage_Regime"] = df[STORAGE_COL].apply(storage_regime)

print("Storage regime thresholds:")
print(f"Low  $\leq$  {q_low:.1f} mm")
print(f"High  $\geq$  {q_high:.1f} mm")

reg_df = test_df.copy().sort_values("YearMonth").reset_index(drop=True)

q05 = reg_df["Storage_pct"].quantile(0.05)
q10 = reg_df["Storage_pct"].quantile(0.10)
q20 = reg_df["Storage_pct"].quantile(0.20)

reg_df["LowStorage_05"] = (reg_df["Storage_pct"] <= q05).astype(int)
reg_df["LowStorage_10"] = (reg_df["Storage_pct"] <= q10).astype(int) # already exists,
keep consistent
reg_df["LowStorage_20"] = (reg_df["Storage_pct"] <= q20).astype(int)

reg_df[["YearMonth", "Storage_pct", "LowStorage_05", "LowStorage_10", "LowStorage_20"]].head(),
(q05, q10, q20)

final_cols = [
    "YearMonth",
    "Price_mean",
    "Storage_active_mm",
    "Storage_pct",
    "RiverFlow_mean",
    "TributaryFlow_mean",
    "Inflow_total",
    "Inflow_z",
    "HVDC_mean",
    "Spill_mean_mm",
    "LowStorage_10",
    "Price_lag1",
    "Storage_pct_lag1",
    "Inflow_z_lag1",
    "LogPrice"
]

final_df = test_df[final_cols].copy()
final_df.head()

# verify lag construction
(final_df[["Price_mean", "Price_lag1"]].head(5),
 final_df[["Inflow_z", "Inflow_z_lag1"]].head(5))

ew_df = final_df.copy().sort_values("YearMonth").reset_index(drop=True)

# Storage momentum
ew_df["Delta_Storage"] = ew_df["Storage_pct"] - ew_df["Storage_pct_lag1"]

```

```

# Rolling inflow stress (3-month mean)
ew_df["Inflow_z_3m"] = ew_df["Inflow_z"].rolling(3).mean()

# Rolling price acceleration (3-month change)
ew_df["Price_3m_change"] = ew_df["Price_mean"] - ew_df["Price_mean"].shift(3)

ew_df = ew_df.dropna().reset_index(drop=True)

ew_df.head()

```

H.3 Regime Detection Code

```

ep = reg_df[["YearMonth", "LowStorage_10", "Price_mean", "Storage_pct"]].copy()

# identify episode starts
ep["episode_start"] = (ep["LowStorage_10"].diff() == 1).astype(int)
ep["episode_id"] = ep["episode_start"].cumsum()

# keep only low-storage months
ep_low = ep[ep["LowStorage_10"] == 1].copy()

episode_summary = (
    ep_low.groupby("episode_id")
        .agg(
            start=("YearMonth", "min"),
            end=("YearMonth", "max"),
            duration_months=("YearMonth", "count"),
            min_storage=("Storage_pct", "min"),
            avg_price=("Price_mean", "mean"),
            max_price=("Price_mean", "max")
        )
        .sort_values(["duration_months", "max_price"], ascending=False)
        .reset_index(drop=True)
)

episode_summary.head(10)

# Transition probabilities (added for Appendix)
reg = final_df.sort_values("YearMonth").copy()
reg["Regime"] = reg["LowStorage_10"].astype(int) # 1 = Low storage, 0 = normal
reg["Regime_next"] = reg["Regime"].shift(-1)
trans = (
    reg.dropna(subset=["Regime_next"])
        .groupby(["Regime", "Regime_next"])
        .size()
        .groupby(level=0)
        .apply(lambda s: s / s.sum())
        .reset_index(name="Probability")
)

trans["From"] = trans["Regime"].map({0: "Normal", 1: "LowStorage_10"})
trans["To"] = trans["Regime_next"].map({0: "Normal", 1: "LowStorage_10"})
trans = trans[["From", "To", "Probability"]].sort_values(["From", "To"])
trans

```


H.4 Model Estimation Code

```
m_drivers = smf.ols(
    "LogPrice ~ Storage_pct + HVDC_mean + Spill_mean_mm + Inflow_z_lag1 + Price_lag1",
    data=test_df
).fit(cov_type="HC3")

m_drivers.summary()

from patsy import dmatrix
import statsmodels.api as sm

# Spline basis for storage
X_spline = dmatrix("bs(Storage_pct, df=5, degree=3, include_intercept=False)", data=reg_df,
    return_type="dataframe")

# Add other covariates
X = pd.concat([
    X_spline,
    reg_df[["HVDC_mean", "Spill_mean_mm", "Inflow_z_lag1", "Price_lag1"]]
], axis=1)

X = sm.add_constant(X)
y = reg_df["LogPrice"].astype(float)

m_spline = sm.OLS(y, X).fit(cov_type="HC3")
m_spline.summary()
```

H.5 Tail-Risk & VaR Code

```
def var_cvar(x, q=0.95):
    x = pd.Series(x).dropna().values
    v = np.percentile(x, 100*q)
    c = x[x >= v].mean() if (x >= v).any() else np.nan
    return v, c

rows = []
for label, subset in [("Normal", tail_df[tail_df["LowStorage_10"]==0]),
    ("LowStorage_10", tail_df[tail_df["LowStorage_10"]==1])]:
    v95, c95 = var_cvar(subset["Price_mean"], 0.95)
    v99, c99 = var_cvar(subset["Price_mean"], 0.99)
    rows.append({
        "Regime": label,
        "n": len(subset),
        "VaR95": v95,
        "CVaR95": c95,
        "VaR99": v99,
        "CVaR99": c99
    })

tail_risk_by_regime = pd.DataFrame(rows)
tail_risk_by_regime

u = P95 # threshold
tail_df["Exceed_u"] = (tail_df["Price_mean"] > u).astype(int)
tail_df["Excess"] = np.where(tail_df["Price_mean"] > u, tail_df["Price_mean"] - u, np.nan)
```

```

exc = tail_df[tail_df["Exceed_u"] == 1].copy()

exceed_summary = exc.groupby("LowStorage_10").agg(
    n_exc=("Exceed_u", "sum"),
    exc_rate=("Exceed_u", "mean"),
    mean_excess=("Excess", "mean"),
    max_excess=("Excess", "max"),
    max_price=("Price_mean", "max")
)

exceed_summary.index = exceed_summary.index.map({0: "Normal", 1: "LowStorage_10"})
exceed_summary

def cond_rate(df, regime_col, event_col):
    g = df.groupby(regime_col)[event_col].mean()
    g.index = g.index.map({0: "Normal", 1: regime_col})
    return g

events = ["Extreme_500", "Severe_1000", "Extreme_P95", "Extreme_P99"]

rows = []
for e in events:
    r = cond_rate(tail_df, "LowStorage_10", e)
    rows.append(pd.DataFrame({"Event": e, "Normal": r.get("Normal", np.nan),
                              "LowStorage_10": r.get("LowStorage_10", np.nan)}, index=[0]))

tail_event_rates = pd.concat(rows, ignore_index=True)
tail_event_rates

rows = []
for col, label in [("LowStorage_10", "1-in-10 proxy (Low10)",
                    ("LowStorage_05", "1-in-20 proxy (Low05)"))]:
    sub = tail_df[tail_df[col] == 1]
    rows.append({
        "DryYearProxy": label,
        "n_months": len(sub),
        "AvgPrice": sub["Price_mean"].mean(),
        "MedianPrice": sub["Price_mean"].median(),
        "P90": sub["Price_mean"].quantile(0.90),
        "P95": sub["Price_mean"].quantile(0.95),
        "Max": sub["Price_mean"].max(),
        "P(price>500)": (sub["Price_mean"] > 500).mean(),
        "P(price>1000)": (sub["Price_mean"] > 1000).mean()
    })

dry_year_risk_table = pd.DataFrame(rows)
dry_year_risk_table

```

H.6 Forecast & Early Warning Code

```
fc_df = final_df.copy().sort_values("YearMonth").reset_index(drop=True)

H = 3 # forecast horizon (months)

# Future dry-regime entry
fc_df["DryNext_H"] = (
    fc_df["Storage_pct"]
    .shift(-1)
    .rolling(H)
    .min()
    .le(p10_storage)
    .astype(int)
)

# Future tail-price event
price_p95 = fc_df["Price_mean"].quantile(0.95)

fc_df["TailNext_H"] = (
    fc_df["Price_mean"]
    .shift(-1)
    .rolling(H)
    .max()
    .ge(price_p95)
    .astype(int)
)

fc_df = fc_df.dropna().reset_index(drop=True)

fc_df[["YearMonth", "DryNext_H", "TailNext_H"]].head()

fc_features = [
    "Storage_pct",
    "Storage_pct_lag1",
    "Inflow_z",
    "Inflow_z_lag1",
    "HVDC_mean",
    "Price_lag1"
]

X_fc = sm.add_constant(fc_df[fc_features])

y_dry = fc_df["DryNext_H"]

dry_fc_model = sm.Logit(y_dry, X_fc).fit(dispatch=False)
dry_fc_model.summary()

# Use same X_fc you created earlier
# Ensure no missing values
X_tail = X_fc.copy()
mask = y_tail.notna()
X_tail = X_tail.loc[mask]
y_tail2 = y_tail.loc[mask]
```

```

# Regularised Logistic regression (stable under separation)
tail_fc_model = sm.Logit(y_tail2, X_tail).fit_regularized(
    method="l1",      # sparse, stable
    alpha=1.0,        # increase if still unstable (2.0, 5.0)
    disp=False
)

tail_fc_model

fc_df.loc[mask, "P_TailNext_H"] = tail_fc_model.predict(X_tail)

fc_df[["YearMonth", "TailNext_H", "P_TailNext_H"]].tail()

fc_df["P_DryNext_H"] = dry_fc_model.predict(X_fc)
fc_df["P_TailNext_H"] = tail_fc_model.predict(X_fc)

plt.figure(figsize=(10,4))
plt.plot(fc_df["YearMonth"], fc_df["P_DryNext_H"], label="P(Dry regime in next 3m)")
plt.plot(fc_df["YearMonth"], fc_df["P_TailNext_H"], label="P(Tail price in next 3m)")
plt.axhline(0.20, linestyle="--", color="green")
plt.axhline(0.50, linestyle="--", color="red")
plt.ylabel("Probability")
plt.xlabel("Year")
plt.title("Forecast Risk Probabilities (3-Month Horizon)")
plt.legend()
plt.tight_layout()
plt.show()

ew_df = final_df.copy().sort_values("YearMonth").reset_index(drop=True)

# Storage momentum
ew_df["Delta_Storage"] = ew_df["Storage_pct"] - ew_df["Storage_pct_lag1"]

# Rolling inflow stress (3-month mean)
ew_df["Inflow_z_3m"] = ew_df["Inflow_z"].rolling(3).mean()

# Rolling price acceleration (3-month change)
ew_df["Price_3m_change"] = ew_df["Price_mean"] - ew_df["Price_mean"].shift(3)

ew_df = ew_df.dropna().reset_index(drop=True)

ew_df.head()

import statsmodels.api as sm

features = [
    "Storage_pct",
    "Delta_Storage",
    "Inflow_z_3m",
    "HVDC_mean",
    "Price_lag1"
]

X = ew_df[features]
X = sm.add_constant(X)
y = ew_df["HighRiskEvent"]

```

```

ew_model = sm.Logit(y, X).fit(displ=False)
ew_model.summary()

ew_df["RiskProb"] = ew_model.predict(X)

def signal(p):
    if p < 0.20:
        return "Green"
    elif p < 0.50:
        return "Amber"
    else:
        return "Red"

ew_df["Signal"] = ew_df["RiskProb"].apply(signal)

ew_df[["YearMonth", "RiskProb", "Signal"]].tail(10)

plt.figure(figsize=(10,4))
plt.plot(ew_df["YearMonth"], ew_df["RiskProb"], label="P(High Risk)")
plt.axhline(0.20, linestyle="--", color="green", label="Green/Amber")
plt.axhline(0.50, linestyle="--", color="red", label="Amber/Red")
plt.ylabel("Probability")
plt.xlabel("Year")
plt.title("Early Warning Probability of High-Risk Price Events")
plt.legend()
plt.tight_layout()
plt.show()

```

H.7 Wellington Retail Proxy Integration Code

The following code ingests the QSDEP retail series, filters the Wellington region, converts to NZD/MWh, and merges it into the monthly panel to visualise wholesale— *Regime analysis and regression results indicate that once*